Minimum cost flow

Outline

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- The algorithm
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In summary

Learning objectives

At the end of this chapter you will be able to:

- build the residual network of a flow with residual capacities and costs, compute the cost of one of its cycles, and augment the flow along it;
- decompose a circulation, and the difference of two feasible flows, into cycles;
- prove that a feasible flow is optimal if and only if its residual network has no negative cycle, and if and only if potentials with non-negative reduced costs exist, and recognize an unbounded instance;
- find a first feasible flow with a maximum flow;
- run the cycle-canceling algorithm and the successive shortest path algorithm, and prove their correctness.

Prerequisites

- From this module, the chapter «Shortest paths»: the Bellman–Ford algorithm with the detection of a negative cycle, Dijkstra's algorithm, and the potentials $\pi_i = -\tilde{d}(i)$ as a certificate.
- The chapter «Network flow models»: the minimum cost flow model, the flow across a cut, the substitution of the lower bounds, the potentials and the reduced costs.
- The chapter «Maximum flow»: the residual network $G(\boldsymbol{x})$, augmenting paths and the Edmonds–Karp algorithm.

Where we were

- In the chapter «Network flow models» the minimum cost flow problem is the LP

$$\min\{\boldsymbol{c}'\boldsymbol{x} : \boldsymbol{B}\boldsymbol{x} = \boldsymbol{b},\ \boldsymbol{l} \leq \boldsymbol{x} \leq \boldsymbol{u}\}$$

  Potentials π_i whose reduced costs $\bar{c}_{ij} = c_{ij} - \pi_i + \pi_j$ are positive only on empty arcs and negative only on saturated arcs certify that a flow is optimal.

- On the reference network the optimal flow $\tilde{\boldsymbol{x}}$ costs 60, and $\boldsymbol{\pi} = (15, 13, 13, 5, 10, 0)$ certifies it.

Where we were: from the certificate to an algorithm

- In the chapter «Maximum flow» a flow grows along augmenting paths of the residual network $G(\boldsymbol{x})$.
- In this chapter the certificate becomes an algorithm:
  - we lower the cost of a feasible flow along cycles of $G(\boldsymbol{x})$,
  - or we build the flow along shortest paths of $G(\boldsymbol{x})$ keeping the certificate true.

The residual network with costs

Introduction: The residual network with costs

- In this section we add the costs to the residual network of the maximum flow chapter
- A forward arc where the flow can grow, a backward arc where it can decrease, with the opposite cost
- Augmenting along a cycle of this network keeps a flow feasible and changes its cost: it is the basic move of the whole chapter

The network of this chapter

- The network $G = (N, A)$ has
  - balances b_i with $\sum_{i \in N} b_i = 0$,
  - lower bounds l_{ij},
  - capacities u_{ij} (finite or $+\infty$),
  - costs c_{ij},

  as in the chapter «Network flow models».

- The arcs are numbered $a_1, a_2, \ldots, a_m$ in lexicographic order.

No pair of opposite arcs

- The set A has no pair of opposite arcs (i, j), (j, i).
- An arc (j, i) opposite to (i, j) can be replaced by the arcs (j, k) and (k, i) through a new transshipment node k:
  - both with the bounds of (j, i),
  - the first with its cost and the second with cost 0.
- The flows and their costs do not change.

Definition 1: residual network

Let $\boldsymbol{x}$ *be a vector with* $l_{ij} \leq x_{ij} \leq u_{ij}$ *for every arc. The* ***residual network*** $G(\boldsymbol{x}) = (N, A(\boldsymbol{x}))$ *has, for every arc* $(i, j) \in A$*:*

- *the* ***forward arc*** (i, j)*, if* $x_{ij} < u_{ij}$*, with* ***residual capacity*** $r_{ij} = u_{ij} - x_{ij}$ *and cost* c_{ij}*;*
- *the* ***backward arc*** (j, i)*, if* $x_{ij} > l_{ij}$*, with residual capacity* $r_{ji} = x_{ij} - l_{ij}$ *and cost* $-c_{ij}$*.*

We write c_{ij}^{x} *for the cost of the arc* (i, j) *of* $G(\boldsymbol{x})$*.*

Forward and backward arcs

- A forward arc can carry r_{ij} more units on (i, j).
- A backward arc (j, i) sends units back from j to i by decreasing x_{ij}, and saves c_{ij} per unit.
- An arc of A gives two arcs of $G(\boldsymbol{x})$ exactly when $l_{ij} < x_{ij} < u_{ij}$.
- With $\boldsymbol{l} = 0$ and without the costs, $G(\boldsymbol{x})$ is the residual network of the chapter «Maximum flow».

Definition 2: cycles of the residual network

Let W be a directed cycle of $G(\boldsymbol{x})$. Its ***cost*** *and its* ***residual capacity*** *are*

$$c(W) = \sum_{(i,j)\in W} c_{ij}^{x}, \qquad r(W) = \min_{(i,j)\in W} r_{ij},$$

and W is a ***negative cycle*** *if $c(W) < 0$.*

Augmenting x *along W by θ, with $0 < \theta \leq r(W)$, means*

- *increasing x_{ij} by θ for every forward arc (i,j) of W,*
- *decreasing x_{ij} by θ for every arc $(i,j) \in A$ whose backward arc (j,i) is in W.*

What happens to x when we augment it along a cycle of $G(x)$?

The bounds and the balances still hold,
and the cost changes by θ times the cost of the cycle.

Proposition 1: augmenting along a cycle

Let $\boldsymbol{l} \leq \boldsymbol{x} \leq \boldsymbol{u}$, *let* W *be a cycle of* $G(\boldsymbol{x})$ *and* $0 < \theta \leq r(W)$, *and let* $\boldsymbol{x}'$ *be* $\boldsymbol{x}$ *augmented along* W *by* θ. *Then* $\boldsymbol{l} \leq \boldsymbol{x}' \leq \boldsymbol{u}$, *every node has the same outflow minus inflow in* $\boldsymbol{x}'$ *and in* $\boldsymbol{x}$, *and*

$$\boldsymbol{c}'\boldsymbol{x}' = \boldsymbol{c}'\boldsymbol{x} + \theta\, c(W).$$

Proof (1/2)

We check the three claims arc by arc.

1. *Bounds.* On a forward arc,

$$x'_{ij} = x_{ij} + \theta \leq x_{ij} + r_{ij} = u_{ij}.$$

   If the backward arc (j, i) is in W, then

$$x'_{ij} = x_{ij} - \theta \geq x_{ij} - r_{ji} = l_{ij}.$$

2. *Balances.* Augmenting by θ along an arc (i, k) of $G(\boldsymbol{x})$ raises the outflow minus inflow of i by θ and lowers that of k by θ: a forward arc sends more flow from i to k, a backward arc sends less flow from k to i. Every node of W is the tail of one arc of W and the head of another one, so the changes cancel.

Proof (2/2)

3. *Cost.* A forward arc changes the cost by

$$c_{ij}\theta = c^x_{ij}\theta;$$

a backward arc (j, i) changes it by

$$-c_{ij}\theta = c^x_{ji}\theta.$$

Adding up along W gives $\theta\, c(W)$.

□

Augmenting along a negative cycle

- If $\boldsymbol{x}$ is a feasible flow, so is $\boldsymbol{x}'$.
- Along a negative cycle the cost decreases by $\theta\,|c(W)|$.

Next: the residual network of a flow of the reference network, and a negative cycle in it.

Example: the most expensive flow (1/5)

We use the reference network of the chapter «Network flow models», with zero lower bounds and the balances

$$\boldsymbol{b} = (5, 2, 0, -3, 0, -4).$$

The costs and the capacities are

arc	$(1,2)$	$(1,3)$	$(2,4)$	$(3,2)$	$(3,5)$	$(4,3)$	$(4,6)$	$(5,4)$	$(5,6)$
c_{ij}	2	2	5	2	3	3	5	5	3
u_{ij}	4	4	4	2	4	3	5	3	2

Example: the most expensive flow (2/5)

- The most expensive feasible flow, computed with Gurobi by maximizing the cost, is

$$\boldsymbol{x}^0 = (1, 4, 4, 1, 4, 1, 3, 3, 1), \qquad \boldsymbol{c}'\boldsymbol{x}^0 = 80,$$

  with the entries in the order of the arcs.

- It is the only flow of cost 80, as the enumeration of the 12 integer feasible flows confirms.

- Each arc of A gives one or two arcs of $G(\boldsymbol{x}^0)$, with their residual capacity and cost: they are in the table of the next slide.

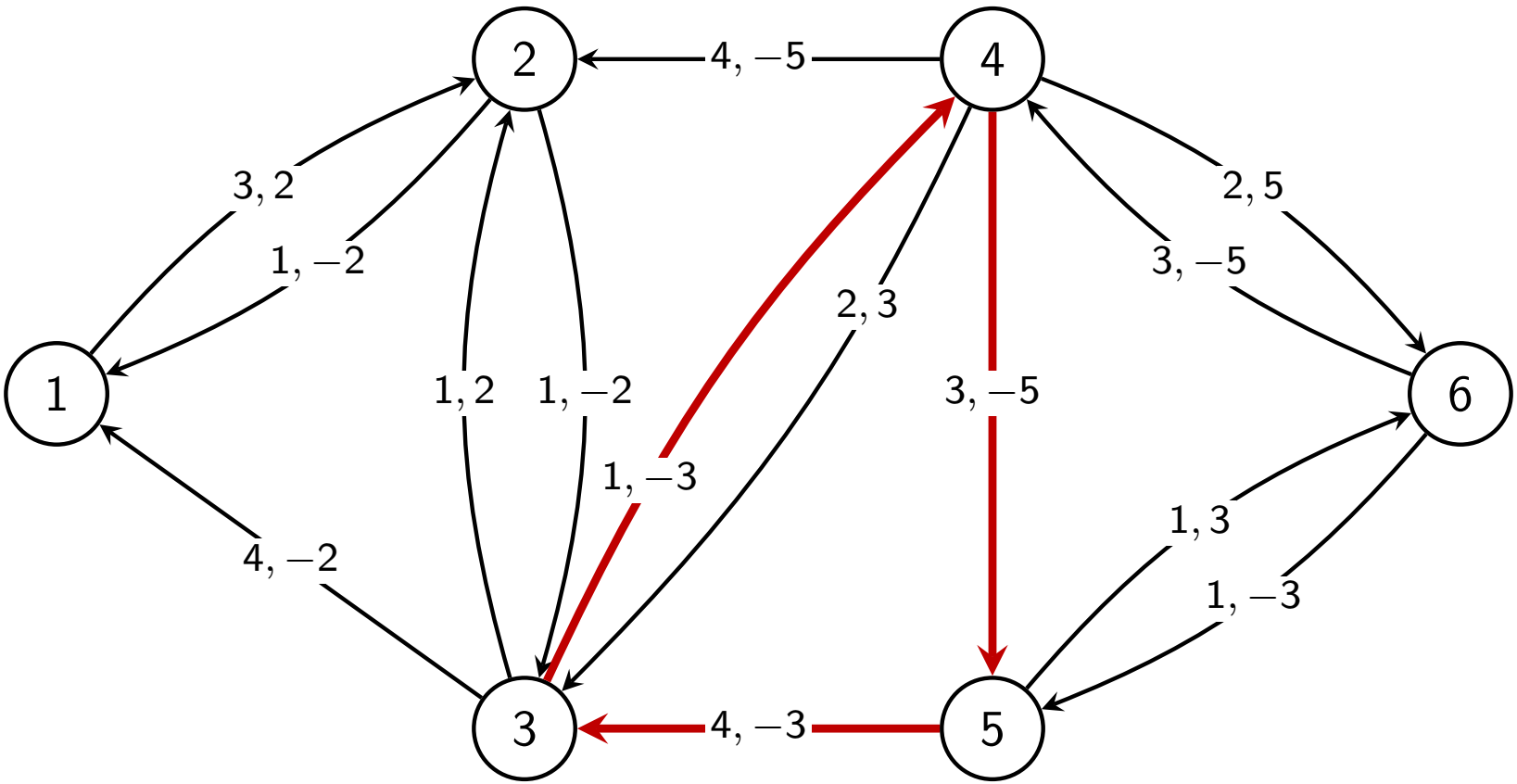
Example: the most expensive flow (3/5)

Each arc gives the following arcs of $G(\boldsymbol{x}^0)$; each entry is the pair r, c of residual capacity and cost:

arc (i,j)	$(1,2)$	$(1,3)$	$(2,4)$	$(3,2)$	$(3,5)$	$(4,3)$	$(4,6)$	$(5,4)$	$(5,6)$
x_{ij}^0	1	4	4	1	4	1	3	3	1
forward (i,j)	$3,2$	–	–	$1,2$	–	$2,3$	$2,5$	–	$1,3$
backward (j,i)	$1,-2$	$4,-2$	$4,-5$	$1,-2$	$4,-3$	$1,-3$	$3,-5$	$3,-5$	$1,-3$

Example: the most expensive flow (4/5)

Each arc of $G(\boldsymbol{x}^0)$ carries the pair r_{ij}, c_{ij}^x; the cycle $W = 4, 5, 3, 4$ is red.



Example: the most expensive flow (5/5)

- The cycle $W = 4, 5, 3, 4$ uses the backward arcs of $(5, 4)$, $(3, 5)$ and $(4, 3)$:

$$c(W) = -5 - 3 - 3 = -11, \qquad r(W) = \min\{3, 4, 1\} = 1.$$

- Augmenting by $\theta = 1$ decreases x_{54}, x_{35} and x_{43} by one unit. The new flow

$$\boldsymbol{x}^1 = (1, 4, 4, 1, 3, 0, 3, 2, 1)$$

  is feasible and costs $80 - 11 = 69$.

- The unit that went around the directed cycle $3, 5, 4, 3$ of G cost 11 and delivered nothing.

Summary

The residual network $G(\boldsymbol{x})$:

- a forward arc of cost c_{ij} where x_{ij} can grow, with $r_{ij} = u_{ij} - x_{ij}$
- a backward arc of cost $-c_{ij}$ where x_{ij} can decrease, with $r_{ji} = x_{ij} - l_{ij}$

Augmenting along a cycle W of $G(\boldsymbol{x})$:

- keeps the bounds and the balances
- changes the cost by $\theta\, c(W)$: along a negative cycle the cost decreases

Circulations and their decomposition

Introduction: Circulations and their decomposition

- In this section we study the vectors that are balanced at every node: the circulations
- Every circulation is a sum of positive multiples of directed cycles
- The difference of two feasible flows is a circulation on $G(\boldsymbol{x})$: every feasible flow is reached from $\boldsymbol{x}$ along cycles of $G(\boldsymbol{x})$, and this is the key of the optimality conditions

Definition 3: circulation

*Let $H = (N, F)$ be a directed graph. A **circulation** on H is a vector $\boldsymbol{y} \in \mathbb{R}^F$ with $\boldsymbol{y} \geq 0$ such that every node has outflow equal to inflow:*

$$\sum_{(i,j)\in\delta^+(i)} y_{ij} = \sum_{(j,i)\in\delta^-(i)} y_{ji}, \qquad \forall i \in N.$$

*The **incidence vector** $\boldsymbol{\chi}^W$ of a directed cycle W of H, with $\chi^W_a = 1$ on the arcs of W and $\chi^W_a = 0$ elsewhere, is a circulation.*

Lemma 1: decomposition into cycles

Let $\boldsymbol{y} \neq 0$ be a circulation on $H = (N, F)$, with p arcs where $y_a > 0$. Then there are directed cycles $W_1, W_2, \ldots, W_k$ of H, with $k \leq p$, made of arcs with $y_a > 0$, and numbers $\theta_1, \theta_2, \ldots, \theta_k > 0$ such that

$$\boldsymbol{y} = \sum_{h=1}^{k} \theta_h \boldsymbol{\chi}^{W_h}.$$

If $\boldsymbol{y}$ is integer, the numbers θ_h are integers.

The proof builds the cycles one at a time:

follow the arcs with $y > 0$ until a node repeats,
remove the cycle found, and repeat.

Proof (1/3)

By induction on p.

1. *A cycle.* Let (i_0, i_1) be an arc with $y_{i_0 i_1} > 0$.
   - Node i_1 receives a positive inflow, so by the balance some arc (i_1, i_2) has $y_{i_1 i_2} > 0$.
   - Going on in the same way we build a directed walk of arcs with $y > 0$.
   - Since N is finite a node repeats, and the piece of the walk between its two occurrences is a directed cycle W_1.

Proof (2/3)

2. *Removing it.* Let $\theta_1 = \min_{a \in W_1} y_a > 0$. The vector

$$\boldsymbol{y}' = \boldsymbol{y} - \theta_1 \boldsymbol{\chi}^{W_1}$$

- is non-negative,
- is balanced, as a difference of balanced vectors,
- has at most $p - 1$ positive arcs, since $y'_a = 0$ on an arc of W_1 that attains the minimum.

Proof (3/3)

3. *Induction.* If $\boldsymbol{y}' = 0$, then $\boldsymbol{y} = \theta_1 \boldsymbol{\chi}^{W_1}$. Otherwise the induction gives

$$\boldsymbol{y}' = \sum_{h=2}^{k} \theta_h \boldsymbol{\chi}^{W_h}$$

with $k - 1 \leq p - 1$ cycles of arcs with $y'_a > 0$, hence $y_a > 0$.

If $\boldsymbol{y}$ is integer, θ_1 is one of its entries and $\boldsymbol{y}'$ is integer, so all the θ_h are integers.

□

Lemma 2: the difference of two feasible flows

Let $\boldsymbol{x}$ and $\boldsymbol{x}'$ be feasible flows. Define $\boldsymbol{y}$ on the arcs of $G(\boldsymbol{x})$: for every $(i,j) \in A$,

$$y_{ij} = x'_{ij} - x_{ij} \quad \text{on the forward arc } (i,j), \text{ if } x'_{ij} > x_{ij},$$
$$y_{ji} = x_{ij} - x'_{ij} \quad \text{on the backward arc } (j,i), \text{ if } x'_{ij} < x_{ij},$$

and $y = 0$ on the other arcs of $G(\boldsymbol{x})$. Then $\boldsymbol{y}$ is a circulation on $G(\boldsymbol{x})$ with $y_{ij} \leq r_{ij}$, and

$$\boldsymbol{c}'\boldsymbol{x}' - \boldsymbol{c}'\boldsymbol{x} = \sum_{(i,j)\in A(\boldsymbol{x})} c^x_{ij}\, y_{ij} = \sum_{h=1}^{k} \theta_h\, c(W_h)$$

for every decomposition $\boldsymbol{y} = \sum_{h=1}^{k} \theta_h \boldsymbol{\chi}^{W_h}$ into cycles of $G(\boldsymbol{x})$.

Proof (1/3)

We check that $\boldsymbol{y}$ lives on $G(\boldsymbol{x})$, that it is balanced, and its cost.

1. If $x'_{ij} > x_{ij}$, then $x_{ij} < x'_{ij} \leq u_{ij}$: the forward arc exists and

$$y_{ij} \leq u_{ij} - x_{ij} = r_{ij}.$$

If $x'_{ij} < x_{ij}$, then $x_{ij} > x'_{ij} \geq l_{ij}$: the backward arc exists and

$$y_{ji} \leq x_{ij} - l_{ij} = r_{ji}.$$

Proof (2/3)

2. As in the proof of Proposition 1, y on an arc (i, k) of $G(\boldsymbol{x})$ adds y to the outflow minus inflow of i and subtracts it from that of k, exactly as the change from $\boldsymbol{x}$ to $\boldsymbol{x}'$ does on the corresponding arc of A.

   Hence the outflow minus inflow of $\boldsymbol{y}$ at node i is

$$(\boldsymbol{B}\boldsymbol{x}')_i - (\boldsymbol{B}\boldsymbol{x})_i = b_i - b_i = 0.$$

Proof (3/3)

3. A forward arc contributes $c_{ij}(x'_{ij} - x_{ij})$, a backward arc

$$-c_{ij}(x_{ij} - x'_{ij}) = c_{ij}(x'_{ij} - x_{ij}) :$$

the total is $\boldsymbol{c}'(\boldsymbol{x}' - \boldsymbol{x})$. With the decomposition,

$$\sum c^x_{ij} y_{ij} = \sum_h \theta_h \sum_{(i,j)\in W_h} c^x_{ij} = \sum_h \theta_h \, c(W_h).$$

□

Every feasible flow is reached along cycles

- By Lemma 1 such a decomposition exists: $\boldsymbol{x}'$ is $\boldsymbol{x}$ augmented along the cycles $W_1, W_2, \ldots, W_k$ of $G(\boldsymbol{x})$ by $\theta_1, \theta_2, \ldots, \theta_k$.
- Since $\boldsymbol{y}$ is positive on at most one arc of $G(\boldsymbol{x})$ per arc of A, $k \leq m$.

Next: the decomposition on the reference network, from the most expensive flow to the optimal flow.

Example: from $\boldsymbol{x}^0$ to $\tilde{\boldsymbol{x}}$ (1/2)

- We compare $\boldsymbol{x}^0$ of the first example, of cost 80, with the optimal flow of the chapter «Network flow models», of cost 60:

$$\tilde{\boldsymbol{x}} = (2, 3, 4, 0, 3, 0, 2, 1, 2).$$

- The difference is

$$\tilde{\boldsymbol{x}} - \boldsymbol{x}^0 = (1, -1, 0, -1, -1, -1, -1, -2, 1),$$

so $\boldsymbol{y}$ is positive on eight arcs of $G(\boldsymbol{x}^0)$: the forward arcs $(1, 2)$ and $(5, 6)$, with $y = 1$, and the backward arcs

$$y_{31} = 1, \quad y_{23} = 1, \quad y_{53} = 1, \quad y_{34} = 1, \quad y_{64} = 1, \quad y_{45} = 2.$$

Example: from $\boldsymbol{x}^0$ to $\tilde{\boldsymbol{x}}$ (2/2)

We follow the proof of Lemma 1, starting each time from the first arc with $y > 0$ in lexicographic order and leaving every node along its arc with $y > 0$ of smallest head:

h	cycle W_h	θ_h	$c(W_h)$
1	$1, 2, 3, 1$	1	$2 - 2 - 2 = -2$
2	$3, 4, 5, 3$	1	$-3 - 5 - 3 = -11$
3	$4, 5, 6, 4$	1	$-5 + 3 - 5 = -7$

- By Lemma 2, $\boldsymbol{c}'\tilde{\boldsymbol{x}} - \boldsymbol{c}'\boldsymbol{x}^0 = -2 - 11 - 7 = -20 = 60 - 80$.
- All three cycles are negative, and W_2 is the cycle $4, 5, 3, 4$ of the first example, written from node 3.

Summary

Circulations:

- A circulation is a sum of positive multiples of directed cycles
- With integer entries, the multiples are integers

The difference of two feasible flows:

- It is a circulation on $G(\boldsymbol{x})$: every feasible flow is reached from $\boldsymbol{x}$ along cycles of $G(\boldsymbol{x})$
- The cost changes by $\sum_h \theta_h \, c(W_h)$

Negative cycle optimality conditions

Introduction: Negative cycle optimality conditions

- In this section we prove the first optimality condition: a feasible flow is optimal exactly when its residual network has no negative cycle
- The proof uses only the decomposition of the previous section
- We also see when the model is unbounded: a negative cycle with infinite capacities

Theorem 1: negative cycle optimality

A feasible flow $\boldsymbol{x}$ is optimal if and only if its residual network $G(\boldsymbol{x})$ has no negative cycle.

Proof (1/2)

We prove the two implications.

A negative cycle improves $\boldsymbol{x}$.

- Let W be a negative cycle of $G(\boldsymbol{x})$ and $\theta = \min\{r(W), 1\} > 0$.
- By Proposition 1, $\boldsymbol{x}$ augmented along W by θ is a feasible flow of cost

$$\boldsymbol{c}'\boldsymbol{x} + \theta\, c(W) < \boldsymbol{c}'\boldsymbol{x},$$

so $\boldsymbol{x}$ is not optimal.

Proof (2/2)

Without negative cycles $\boldsymbol{x}$ is optimal.

- Let $\boldsymbol{x}'$ be any feasible flow. By Lemma 2,

$$\boldsymbol{c}'\boldsymbol{x}' = \boldsymbol{c}'\boldsymbol{x} + \sum_h \theta_h \, c(W_h)$$

  with cycles W_h of $G(\boldsymbol{x})$ and $\theta_h > 0$.
- Every $c(W_h)$ is non-negative, so $\boldsymbol{c}'\boldsymbol{x}' \geq \boldsymbol{c}'\boldsymbol{x}$.

□

Two remarks on the theorem

- The proof compares two flows through their difference, and uses no LP duality.
- When some capacities are infinite the model can be unbounded, and the next proposition tells exactly when.

Proposition 2: unbounded instances

Assume that the MCF *model has a feasible flow. It is unbounded if and only if G has a directed cycle C of negative cost,*

$$\sum_{(i,j)\in C} c_{ij} < 0,$$

whose arcs all have $u_{ij} = +\infty$.

Proof (1/3)

If. Let $\boldsymbol{x}$ be feasible.

- Every arc of C has $x_{ij} < u_{ij} = +\infty$, so C is a cycle of forward arcs of $G(\boldsymbol{x})$ with $r(C) = +\infty$.

- Augmenting along C by any $\theta > 0$ gives feasible flows of cost

$$\boldsymbol{c}'\boldsymbol{x} + \theta\, c(C),$$

which tends to $-\infty$.

Proof (2/3)

Only if. Assume that there is no such cycle C, and fix a feasible flow $\boldsymbol{x}$.

- Let R be the largest finite residual capacity of an arc of $G(\boldsymbol{x})$ ($R = 0$ if there is none), and

$$K = \max_{(i,j) \in A} |c_{ij}|.$$

- Let $\boldsymbol{x}'$ be any feasible flow, decomposed as in Lemma 2.

Proof (3/3)

1. A cycle W_h with $c(W_h) < 0$ has an arc a of finite residual capacity: backward arcs have finite residual capacity, and a cycle of forward arcs with infinite capacity would be a cycle C. Hence
$$\theta_h \leq y_a \leq r_a \leq R.$$
2. W_h has at most n arcs, so $c(W_h) \geq -nK$; there are at most m cycles.

Hence $\boldsymbol{c}'\boldsymbol{x}' \geq \boldsymbol{c}'\boldsymbol{x} - m\,R\,n\,K$ for every feasible $\boldsymbol{x}'$: the model is bounded.

□

Bounded models have an optimal flow

- A feasible LP that is bounded has an optimal solution (Linear Programming, chapter «The geometry of Linear Programming»).
- With finite capacities the MCF model is never unbounded.

Example: a flow that is not optimal (1/2)

- The feasible flow

$$\bar{\boldsymbol{x}} = (1, 4, 3, 0, 4, 0, 2, 2, 2)$$

of the chapter «Network flow models» costs 63.

- Its residual network has the cycle $W = 1, 2, 4, 5, 3, 1$, made of the forward arcs $(1, 2)$, $(2, 4)$ and of the backward arcs $(4, 5)$, $(5, 3)$, $(3, 1)$:

$$c(W) = 2+5-5-3-2 = -3, \quad r(W) = \min\{3, 1, 2, 4, 4\} = 1.$$

Example: a flow that is not optimal (2/2)

- By Theorem 1 $\bar{\boldsymbol{x}}$ is not optimal.
- Augmenting by 1 gives $\tilde{\boldsymbol{x}}$, of cost $63 - 3 = 60$.
- The decrease 3 is the gap of $\bar{\boldsymbol{x}}$ computed with the dual solution in the chapter «Network flow models».

Example: an unbounded instance

- We change the reference network: $c_{54} = -12$ and $u_{35} = u_{54} = u_{43} = +\infty$.
- The directed cycle $3, 5, 4, 3$ of G costs

$$3 - 12 + 3 = -6$$

  and has infinite capacities.
- By Proposition 2 the model is unbounded, as Gurobi confirms: each unit sent around the cycle lowers the cost by 6.

Summary

Optimality:

- A feasible flow is optimal if and only if its residual network has no negative cycle
- The proof uses the decomposition into cycles, not LP duality

Unbounded instances:

- A feasible model is unbounded if and only if a negative directed cycle has all its capacities infinite
- With finite capacities the model is never unbounded

Optimality conditions with potentials

Introduction: Optimality conditions with potentials

- In this section we translate «no negative cycle» into potentials with non-negative reduced costs
- The potentials are a certificate that is easy to check, arc by arc
- On the arcs of A they give the complementary slackness conditions of the chapter «Network flow models», now proved without LP duality

Reduced costs on the residual network

- Given potentials $\boldsymbol{\pi} \in \mathbb{R}^n$, the reduced cost of an arc (i, j) of $G(\boldsymbol{x})$ is

$$c_{ij}^x - \pi_i + \pi_j.$$

- For a forward arc it is $\bar{c}_{ij} = c_{ij} - \pi_i + \pi_j$, the reduced cost of the chapter «Network flow models».
- For the backward arc (j, i) of (i, j) it is

$$-c_{ij} - \pi_j + \pi_i = -\bar{c}_{ij}.$$

The potentials telescope along a cycle

- Along a cycle W of $G(\boldsymbol{x})$ the potentials telescope, as in the chapter «Shortest paths»:

$$c(W) = \sum_{(i,j)\in W} \left(c_{ij}^{x} - \pi_i + \pi_j\right).$$

- So, if all the reduced costs are non-negative, no cycle is negative.

Theorem 2: reduced cost optimality

A feasible flow $\boldsymbol{x}$ *is optimal if and only if there are potentials* $\boldsymbol{\pi} \in \mathbb{R}^n$ *such that*

$$c_{ij}^x - \pi_i + \pi_j \geq 0 \qquad \text{for every arc } (i,j) \text{ of } G(\boldsymbol{x}).$$

Proof (1/2)

If. By the telescoping formula every cycle of $G(\boldsymbol{x})$ has $c(W) \geq 0$, and $\boldsymbol{x}$ is optimal by Theorem 1.

Only if. Let $\boldsymbol{x}$ be optimal: by Theorem 1, $G(\boldsymbol{x})$ has no negative cycle.

1. Add a node r and an arc (r, i) of cost 0 for every $i \in N$. In the new graph every node is reachable from r, and the cycles are those of $G(\boldsymbol{x})$, since no arc enters r: none is negative.

Proof (2/2)

2. By Theorem «Bellman's equations» of the chapter «Shortest paths», the distances $d(i)$ from r are finite and

$$d(j) \leq d(i) + c_{ij}^x \qquad \text{for every arc } (i,j) \text{ of } G(\boldsymbol{x}).$$

3. With $\pi_i = -d(i)$ we get

$$c_{ij}^x - \pi_i + \pi_j = c_{ij}^x + d(i) - d(j) \geq 0.$$

□

Corollary 1: complementary slackness

Let $\boldsymbol{x}$ *be a feasible flow and* $\boldsymbol{\pi} \in \mathbb{R}^n$. *The condition of Theorem 2 holds if and only if, for every arc* $(i, j) \in A$,

$$\bar{c}_{ij} > 0 \;\Rightarrow\; x_{ij} = l_{ij}, \qquad\qquad \bar{c}_{ij} < 0 \;\Rightarrow\; x_{ij} = u_{ij}.$$

In particular, $l_{ij} < x_{ij} < u_{ij}$ *implies* $\bar{c}_{ij} = 0$.

Proof

- The arc $(i,j) \in A$ gives the forward arc, of reduced cost $\bar{c}_{ij}$, when $x_{ij} < u_{ij}$, and the backward arc, of reduced cost $-\bar{c}_{ij}$, when $x_{ij} > l_{ij}$.
- So the condition on the arcs coming from (i,j) reads

$$x_{ij} < u_{ij} \;\Rightarrow\; \bar{c}_{ij} \geq 0, \qquad\qquad x_{ij} > l_{ij} \;\Rightarrow\; \bar{c}_{ij} \leq 0.$$

- Since $l_{ij} \leq x_{ij} \leq u_{ij}$, the first implication is equivalent to $\bar{c}_{ij} < 0 \Rightarrow x_{ij} = u_{ij}$, and the second to $\bar{c}_{ij} > 0 \Rightarrow x_{ij} = l_{ij}$.
- If $l_{ij} < x_{ij} < u_{ij}$, both implications apply and $\bar{c}_{ij} = 0$.



The certificate of the chapter «Network flow models»

- With $\boldsymbol{l} = 0$ these are the conditions of Corollary «reduced cost certificate» of the chapter «Network flow models».
- There they were proved sufficient with LP duality.
- Theorem 2 shows that every optimal flow has such potentials, which with $\alpha_{ij} = \max\{0, -\bar{c}_{ij}\}$ form an optimal dual solution.

The potentials of the proof

- The potentials of the proof are $\pi_i = -d(i)$, minus the distances from r, as $\pi_i = -\tilde{d}(i)$ in the chapter «Shortest paths».
- They can be shifted by a constant; we normalize them with $\pi_n = 0$.

Example: the certificate of the optimal flow (1/2)

We take $\tilde{\boldsymbol{x}}$ and the potentials $\boldsymbol{\pi} = (15, 13, 13, 5, 10, 0)$ of the chapter «Network flow models». The reduced costs of the arcs of $G(\tilde{\boldsymbol{x}})$ are $\bar{c}_{ij}$ on the forward arcs and $-\bar{c}_{ij}$ on the backward arcs:

arc (i,j)	$(1,2)$	$(1,3)$	$(2,4)$	$(3,2)$	$(3,5)$	$(4,3)$	$(4,6)$	$(5,4)$	$(5,6)$
$\tilde{x}_{ij}$	2	3	4	0	3	0	2	1	2
$\bar{c}_{ij}$	0	0	-3	2	0	11	0	0	-7
forward (i,j)	0	0	–	2	0	11	0	0	–
backward (j,i)	0	0	3	–	0	–	0	0	7

Example: the certificate of the optimal flow (2/2)

- All the 14 arcs of $G(\tilde{\boldsymbol{x}})$ have non-negative reduced cost: by Theorem 2, $\tilde{\boldsymbol{x}}$ is optimal.
- In the form of Corollary 1: $(3,2)$ and $(4,3)$ are empty, $(2,4)$ and $(5,6)$ are saturated.
- The potentials are minus the costs of the cheapest walks of $G(\tilde{\boldsymbol{x}})$ ending at each node. For example, the walk $6,4,5,3,1$ uses four backward arcs and costs

$$-5-5-3-2=-15=-\pi_1:$$

in G it is the path $1,3,5,4,6$ of the spanning tree of the chapter «Network flow models».

Summary

Optimality with potentials:

- A feasible flow is optimal if and only if some potentials give non-negative reduced costs on all the arcs of $G(\boldsymbol{x})$
- Such potentials are minus the distances in $G(\boldsymbol{x})$ from a node joined to all the others

On the arcs of A:

- positive $\bar{c}_{ij}$ only on arcs at the lower bound
- negative $\bar{c}_{ij}$ only on saturated arcs

A first feasible flow

Introduction: A first feasible flow

- The cycle-canceling algorithm of the next section improves a feasible flow: it needs one to start from
- In this section we find one with a maximum flow, from a super source to a super sink
- The same computation tells whether the network has a feasible flow at all

Starting from a feasible flow

- An algorithm that improves a feasible flow needs a feasible flow to start from.
- By the substitution of the chapter «Network flow models» we can assume $\boldsymbol{l} = 0$.

Definition 4: auxiliary network

Let $G = (N, A)$ be a balanced network with $\boldsymbol{l} = 0$, and

$$D = \sum_{i \in N : b_i > 0} b_i$$

*its total supply. The **auxiliary network** adds to G*

- *a **super source** s and a **super sink** t,*
- *an arc (s, i) with capacity b_i for every supply node i,*
- *an arc (i, t) with capacity $-b_i$ for every demand node i;*

the arcs of A keep their capacities.

Proposition 3: a feasible flow from a maximum flow

The network G has a feasible flow if and only if the maximum s–t flow of the auxiliary network has value D. In that case, a maximum flow restricted to the arcs of A is a feasible flow of G.

Proof (1/2)

The value of an s–t flow is the flow leaving s, at most the total capacity D of the arcs (s, i).

1. *Only if.* Let $\boldsymbol{x}$ be feasible, and set $x_{si} = b_i$ on the arcs (s, i) and $x_{it} = -b_i$ on the arcs (i, t).
   - At a supply node the outflow minus inflow becomes $b_i - x_{si} = 0$, at a demand node $b_i + x_{it} = 0$, and transshipment nodes do not change.
   - This is an s–t flow of value D, hence maximum.

Proof (2/2)

2. *If.* Let a maximum flow have value D. Then all the arcs (s, i) are saturated. The value is also the flow entering t, and the arcs (i, t) have total capacity

$$\sum_{i:b_i<0} (-b_i) = D,$$

since the network is balanced: they are saturated too.

3. Conservation at $i \in N$ then gives, along the arcs of A, outflow minus inflow equal to

$$x_{si} - x_{it} = b_i,$$

and the capacities hold: the restriction to A is a feasible flow.

□

Computing the maximum flow

- We compute the maximum flow with the Edmonds–Karp algorithm of the chapter «Maximum flow»:
  - a breadth-first search from s in the residual network,
  - scanning the arcs that leave a node by increasing index of the head (s before node 1, t after node n).

- The maximum flow ignores the costs: it gives a feasible flow, usually far from optimal.

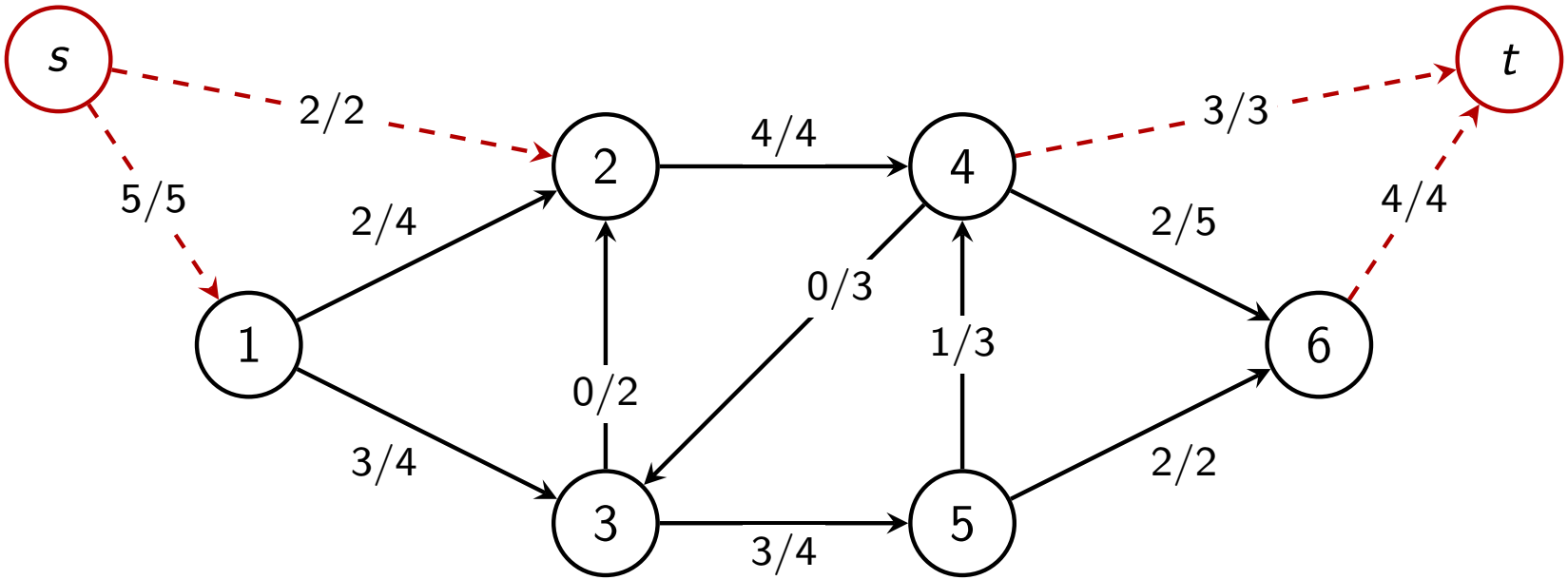
Example: a feasible flow of the reference network (1/2)

- The reference network has $D = 5 + 2 = 7$.
- The auxiliary network adds the arcs $(s, 1)$, $(s, 2)$ with capacities 5, 2, and $(4, t)$, $(6, t)$ with capacities 3, 4.
- The augmenting paths of the Edmonds–Karp algorithm are:

path	nodes	units
1	$s, 2, 4, t$	2
2	$s, 1, 2, 4, t$	1
3	$s, 1, 2, 4, 6, t$	1
4	$s, 1, 3, 5, 6, t$	2
5	$s, 1, 3, 5, 4, 6, t$	1

Example: a feasible flow of the reference network (2/2)

The value is $7 = D$, so the network has a feasible flow. Each arc carries x_{ij}/u_{ij} (the new arcs are red and dashed):



The restriction to A is $(2, 3, 4, 0, 3, 0, 2, 1, 2)$: on this instance it is the optimal flow $\tilde{\boldsymbol{x}}$. This is a coincidence, since the search never looks at the costs.

Summary

The auxiliary network:

- A super source feeds the supply nodes and a super sink is fed by the demand nodes
- Feasibility becomes a maximum flow problem

The test:

- The network has a feasible flow if and only if the maximum flow saturates all the new arcs (value D)
- The maximum flow ignores the costs: its flow is feasible, not optimal in general

The cycle-canceling algorithm

Introduction: The cycle-canceling algorithm

- In this section the negative cycle condition becomes an algorithm: while there is a negative cycle, we augment along it
- We find the negative cycles with the Bellman–Ford algorithm, and at the stop its labels give the potentials
- We prove that the algorithm is correct with integer data, and we run it on the reference network

Canceling a negative cycle

- Theorem 1 suggests an algorithm due to Klein: while $G(\boldsymbol{x})$ has a negative cycle, augment along it.
- Augmenting by $r(W)$ saturates an arc of W, which leaves $G(\boldsymbol{x})$: the cycle is **canceled**.

Finding a negative cycle

- A negative cycle of $G(\boldsymbol{x})$ need not be reachable from a given node.
- We run the Bellman–Ford algorithm of the chapter «Shortest paths» from a new node r joined to every node by an arc of cost 0.
- Its first pass gives label 0 to every node, and we write the table from there.

Proposition 4: Bellman–Ford on the residual network (1/2)

Let $d^0(j) = 0$ *for every* $j \in N$ *and, for* $k \geq 1$,

$$d^k(j) = \min \left\{ d^{k-1}(j), \min_{(i,j) \in A(\boldsymbol{x})} \left(d^{k-1}(i) + c^{\boldsymbol{x}}_{ij} \right) \right\}$$

for every $j \in N$.

(a) $d^k(j)$ *is the minimum cost of a walk of* $G(\boldsymbol{x})$ *with at most* k *arcs that ends at* j, *the walk without arcs having cost* 0.

Proposition: Bellman–Ford on the residual network (2/2)

(b) *$G(\boldsymbol{x})$ has a negative cycle if and only if $d^n \neq d^{n-1}$. In that case, if $d^n(j) < d^{n-1}(j)$, the predecessors read backwards from j, one row per pass from row n to row 1, give a walk of $G(\boldsymbol{x})$ with n arcs, ending at j, that contains a negative cycle.*

(c) *If $d^k = d^{k-1}$ for some $k \leq n$, the potentials $\pi_i = -d^k(i)$ satisfy $c^x_{ij} - \pi_i + \pi_j \geq 0$ on every arc of $G(\boldsymbol{x})$.*

Proof (1/3)

Let H be $G(\boldsymbol{x})$ plus the node r and the arcs (r, i) of cost 0, and let D^k be the labels of Bellman–Ford from r in H:

$$D^1(i) = 0 \quad \text{for every } i \in N, \qquad D^k(r) = 0 \quad \text{for every } k.$$

1. By induction on k, $D^{k+1}(j) = d^k(j)$ for $j \in N$:
   - the recursions coincide, and the arc (r, j) offers $D^k(r) + 0 = 0$, which never lowers the minimum since $d^{k-1}(j) \leq d^0(j) = 0$;
   - a walk from r with $k+1$ arcs is an arc (r, i) followed by a walk of $G(\boldsymbol{x})$ with k arcs from i: this proves (a).

Proof (2/3)

2. In H every node is reachable from r, and the cycles are those of $G(\boldsymbol{x})$.
   - By Theorem «detection of a negative cycle» of the chapter «Shortest paths», applied to the $n+1$ nodes of H, a negative cycle exists if and only if $D^{n+1} \neq D^n$, that is $d^n \neq d^{n-1}$.
   - The walk from r with $n+1$ arcs read in rows $n+1, \ldots, 2$ of H, which are our rows $n, \ldots, 1$, contains a negative cycle.
   - Without its first arc $(r, \cdot)$ it is a walk of $G(\boldsymbol{x})$ with n arcs, and the cycle does not contain r. This proves (b).

Proof (3/3)

3. If $d^k = d^{k-1}$, the next pass repeats the labels:

$$d^k(j) \leq d^k(i) + c_{ij}^x \qquad \text{for every arc } (i,j) \text{ of } G(\boldsymbol{x}),$$

which is (c) with $\pi = -d^k$.

□

Rules of this chapter

- The arcs of $G(\boldsymbol{x})$ are scanned in lexicographic order; as in the chapter «Shortest paths», ties in the minimum go to the smallest tail and a predecessor changes only when the label strictly decreases.
- If $d^n \neq d^{n-1}$, j is the smallest node with $d^n(j) < d^{n-1}(j)$, and the cycle is the piece of the walk between the two occurrences of the first node met twice, reading the walk from its first node.
- At the stop the potentials $\pi_i = -d^k(i)$ are shifted so that $\pi_n = 0$.

Algorithm: Cycle-canceling algorithm

Input : a balanced network with $\boldsymbol{l} = 0$, capacities u_{ij}, costs c_{ij}

Output : an optimal flow $\boldsymbol{x}$ with its certificate $\boldsymbol{\pi}$, or «infeasible», or «unbounded»

1 **Step 0 (start).** If a cycle of arcs with $u_{ij} = +\infty$ has negative cost, `STOP`: unbounded.

2 Find a feasible flow $\boldsymbol{x}$ with a maximum flow of the auxiliary network.

3 If its value is less than D, `STOP`: infeasible.

4 **Step 1 (negative cycle).** Run Bellman–Ford on $G(\boldsymbol{x})$ from $d^0 = 0$.

5 If $d^k = d^{k-1}$ for some $k \leq n$, `STOP`: $\boldsymbol{x}$ is optimal, with $\boldsymbol{\pi} = -d^k$.

6 Otherwise read a negative cycle W from the table.

7 **Step 2 (cancel).** Augment $\boldsymbol{x}$ along W by $\theta = r(W)$ and go to Step 1.

Remarks on Step 0

- The test of Step 0 is Proposition 4 applied to the graph of the arcs with $u_{ij} = +\infty$ and their costs.
- When it fails, every negative cycle of $G(\boldsymbol{x})$ has an arc of finite residual capacity (proof of Proposition 2), so $r(W) < +\infty$ in Step 2.
- Any feasible flow can be used in Step 0; the maximum flow is one way to find it.

Theorem 3: correctness of cycle canceling

Let the balances, the capacities and the costs be integer, with capacities possibly $+\infty$, and let the algorithm pass Step 0 with an integer feasible flow $\boldsymbol{x}^0$. Then

- *every flow of the algorithm is integer and feasible,*
- *every iteration lowers the cost by at least 1,*
- *the algorithm stops after at most $\boldsymbol{c}'\boldsymbol{x}^0 - z(\mathrm{MCF})$ iterations with an optimal flow.*

Proof (1/2)

By Proposition 2 the model is bounded, so it has an optimal value $z(\mathrm{MCF})$.

1. If $\boldsymbol{x}$ is integer, the residual capacities are positive integers or $+\infty$, and $r(W)$ is a positive integer, since $r(W) < +\infty$. By Proposition 1 the new flow is feasible and integer.

2. The cost changes by $r(W)\, c(W)$, where $r(W) \geq 1$ and $c(W) \leq -1$ are integers: it decreases by at least 1.

Proof (2/2)

3. The cost starts at $\boldsymbol{c}'\boldsymbol{x}^0$ and never goes below $z(\mathrm{MCF})$, so there are at most

$$\boldsymbol{c}'\boldsymbol{x}^0 - z(\mathrm{MCF})$$

iterations.

At the stop $G(\boldsymbol{x})$ has no negative cycle (Proposition 4), and $\boldsymbol{x}$ is optimal by Theorem 1.

□

Running time

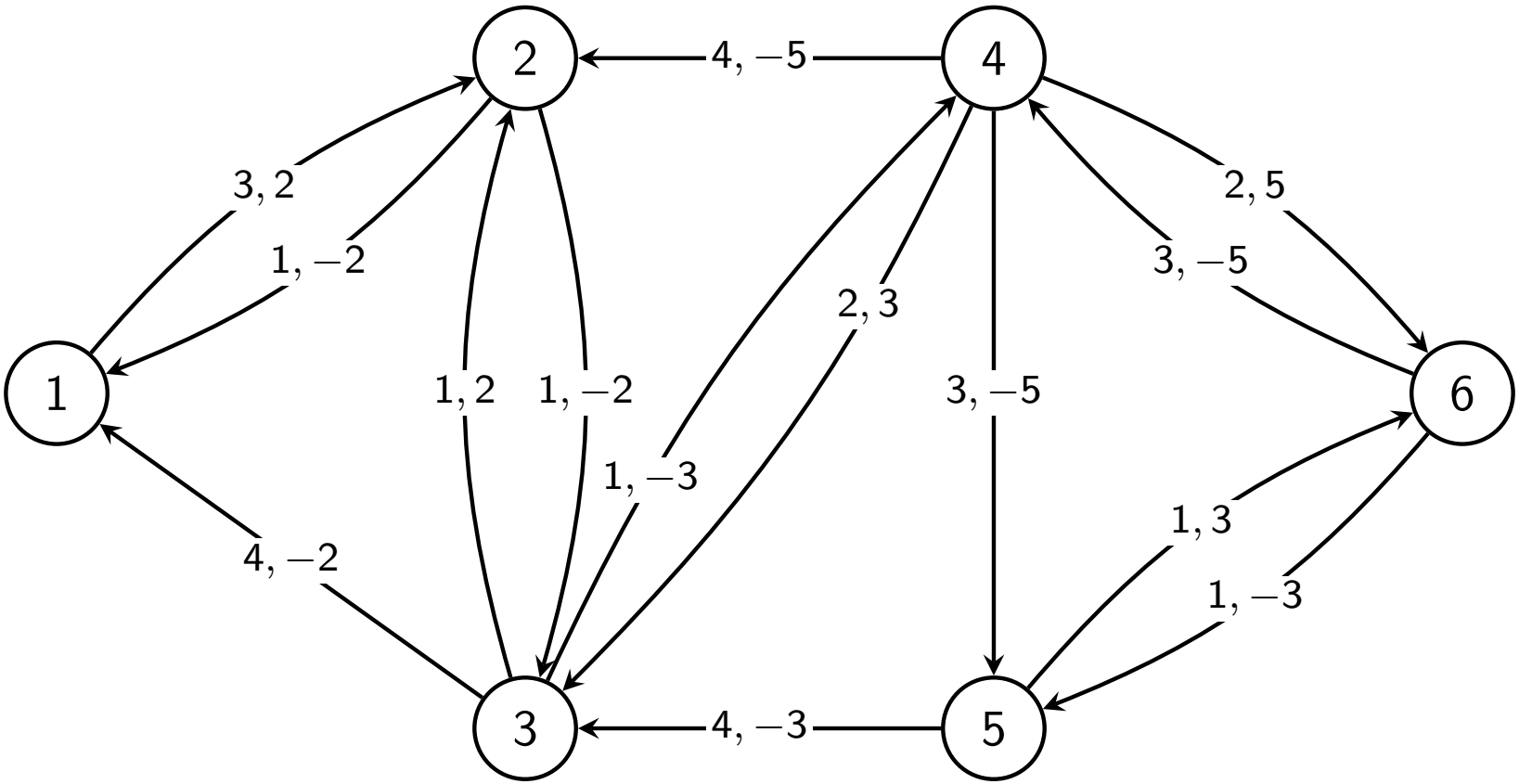
- $G(\boldsymbol{x})$ has at most $2m$ arcs, so Step 1 takes $O(n\,m)$ time.
- With finite capacities, $U = \max u_{ij}$ and $K = \max |c_{ij}|$, every flow costs between $-mKU$ and mKU: at most $2mKU$ iterations, and $O(n\,m^2 KU)$ time.
- The bound depends on the size of the numbers K and U, not only on n and m: it is a **pseudo-polynomial** bound.

Cycle canceling on the reference network

- On the reference network Step 0 returns $\tilde{\boldsymbol{x}}$ (the example of the previous section), and the algorithm stops at the first Bellman–Ford run.
- To see it at work, we start from the most expensive flow $\boldsymbol{x}^0$ of the first example, of cost 80.
- In the tables each row is a pass k; next to a label that decreased, its new predecessor in brackets. The entries read backwards to find the walk are in bold.

Example: iteration 1 (1/4)

$G(\boldsymbol{x}^0)$ is the residual network of the first example, with the pair r_{ij}, c_{ij}^x on each arc:



Example: iteration 1 (2/4)

Bellman–Ford on $G(\boldsymbol{x}^0)$ gives:

k	$d^k(1)$	$d^k(2)$	$d^k(3)$	$d^k(4)$	$d^k(5)$	$d^k(6)$
0	0	0	0	0	0	0
1	-2 (2)	-5 (4)	-3 (5)	-5 (6)	-5 (4)	0
2	-7 (2)	-10 (4)	-8 (5)	-6 (3)	-10 (4)	-2 (5)
3	-12 (2)	-11 (4)	-13 (5)	-11 (3)	-11 (4)	-7 (5)
4	-15 (3)	-16 (4)	-14 (5)	-16 (3)	-16 (4)	-8 (5)
5	-18 (2)	-21 (4)	-19 (5)	-17 (3)	-21 (4)	-13 (5)
6	-23 (2)	-22 (4)	-24 (5)	-22 (3)	-22 (4)	-18 (5)

Example: iteration 1 (3/4)

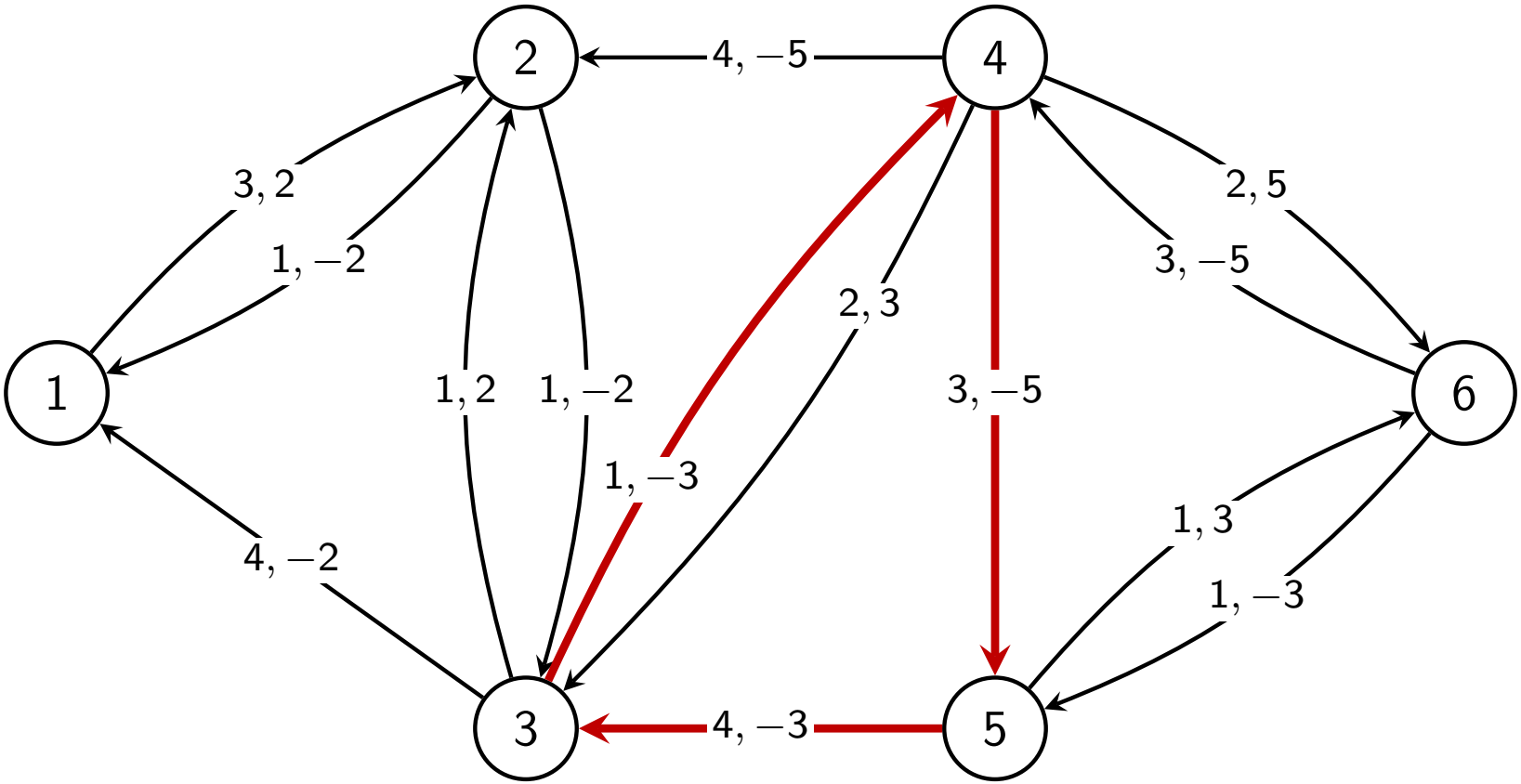
- All the labels decrease in pass 6, so $j = 1$.
- Reading backwards: row 6 gives 2, row 5 for node 2 gives 4, then 3, 5, 4, 6.
- The walk $6, 4, 5, 3, 4, 2, 1$ meets node 4 twice:

$$W = 4, 5, 3, 4, \qquad c(W) = -11, \qquad r(W) = 1.$$

- The new flow is $\boldsymbol{x}^1 = (1, 4, 4, 1, 3, 0, 3, 2, 1)$, of cost 69.

Example: iteration 1 (4/4)

The cycle $W = 4, 5, 3, 4$ in $G(\boldsymbol{x}^0)$ is red: augmenting by $r(W) = 1$ empties the arc $(4, 3)$.



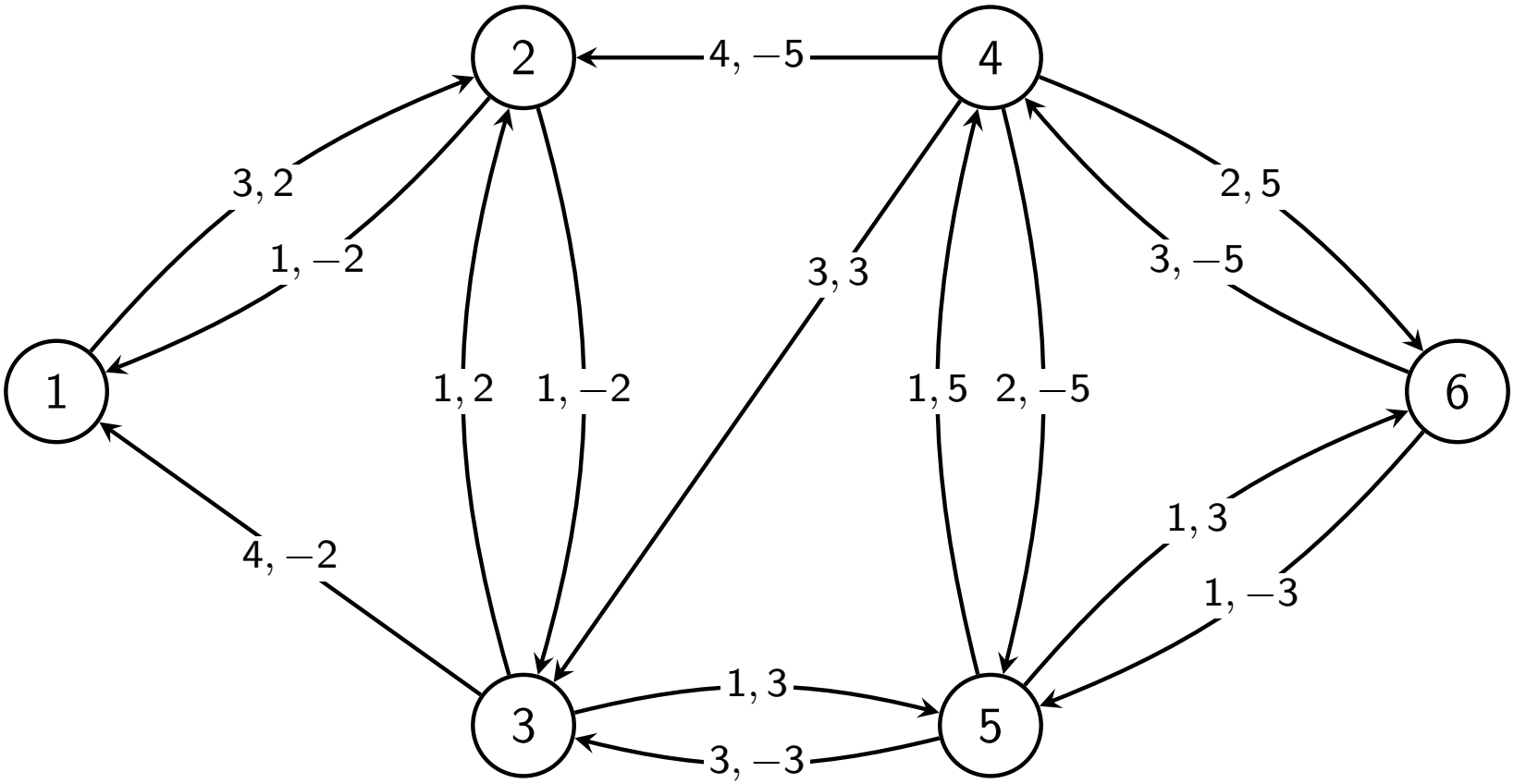
Example: iteration 2 (1/5)

The arcs of W change: $(4,5)$ and $(5,3)$ lose one unit, $(3,4)$ disappears, and the forward arcs $(3,5)$, $(5,4)$ appear. $G(\boldsymbol{x}^1)$ is (each entry is the pair r, c):

arc (i,j)	$(1,2)$	$(1,3)$	$(2,4)$	$(3,2)$	$(3,5)$	$(4,3)$	$(4,6)$	$(5,4)$	$(5,6)$
x_{ij}^1	1	4	4	1	3	0	3	2	1
forward (i,j)	$3,2$	–	–	$1,2$	$1,3$	$3,3$	$2,5$	$1,5$	$1,3$
backward (j,i)	$1,-2$	$4,-2$	$4,-5$	$1,-2$	$3,-3$	–	$3,-5$	$2,-5$	$1,-3$

Example: iteration 2 (2/5)

$G(\boldsymbol{x}^1)$, with the pair r_{ij}, c^x_{ij} on each arc:



Example: iteration 2 (3/5)

Bellman–Ford on $G(\boldsymbol{x}^1)$ gives:

k	$d^k(1)$	$d^k(2)$	$d^k(3)$	$d^k(4)$	$d^k(5)$	$d^k(6)$
0	0	0	0	0	0	0
1	-2 (2)	-5 (4)	-3 (5)	-5 (6)	-5 (4)	0
2	-7 (2)	-10 (4)	-8 (5)	-5	-10 (4)	-2 (5)
3	-12 (2)	-10	-13 (5)	-7 (6)	-10	-7 (5)
4	-15 (3)	-12 (4)	-13	-12 (6)	-12 (4)	-7
5	-15	-17 (4)	-15 (5)	-12	-17 (4)	-9 (5)
6	-19 (2)	-17	-20 (5)	-14 (6)	-17	-14 (5)

Example: iteration 2 (4/5)

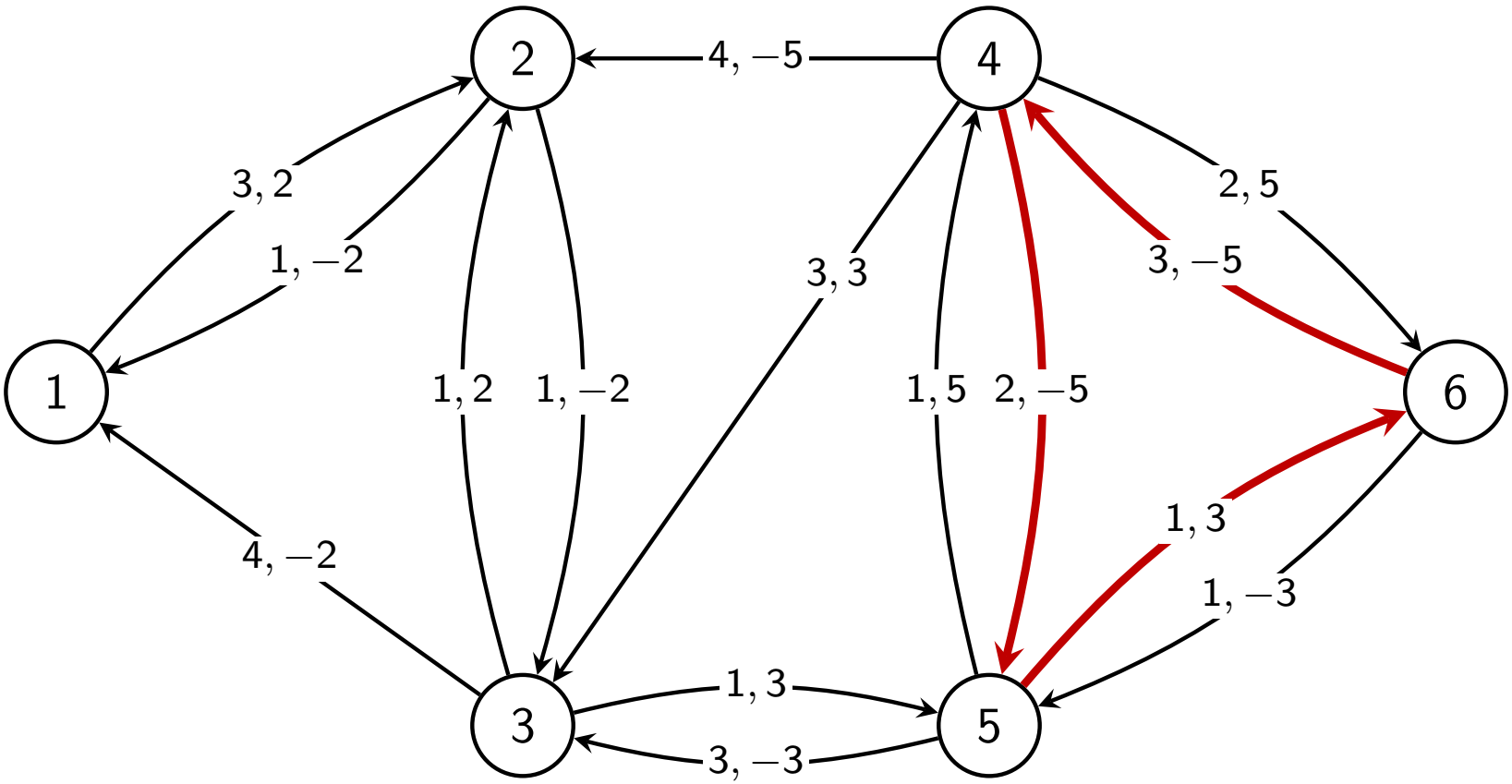
- Again $j = 1$, and the walk is $6, 4, 5, 6, 4, 2, 1$.
- $W = 6, 4, 5, 6$, made of the backward arcs of $(4, 6)$ and $(5, 4)$ and of the forward arc $(5, 6)$:

$$c(W) = -5 - 5 + 3 = -7, \qquad r(W) = \min\{3, 2, 1\} = 1.$$

- The new flow is $\boldsymbol{x}^2 = (1, 4, 4, 1, 3, 0, 2, 1, 2)$, of cost 62.

Example: iteration 2 (5/5)

The cycle $W = 6, 4, 5, 6$ in $G(\boldsymbol{x}^1)$ is red: augmenting by $r(W) = 1$ saturates the arc $(5, 6)$.

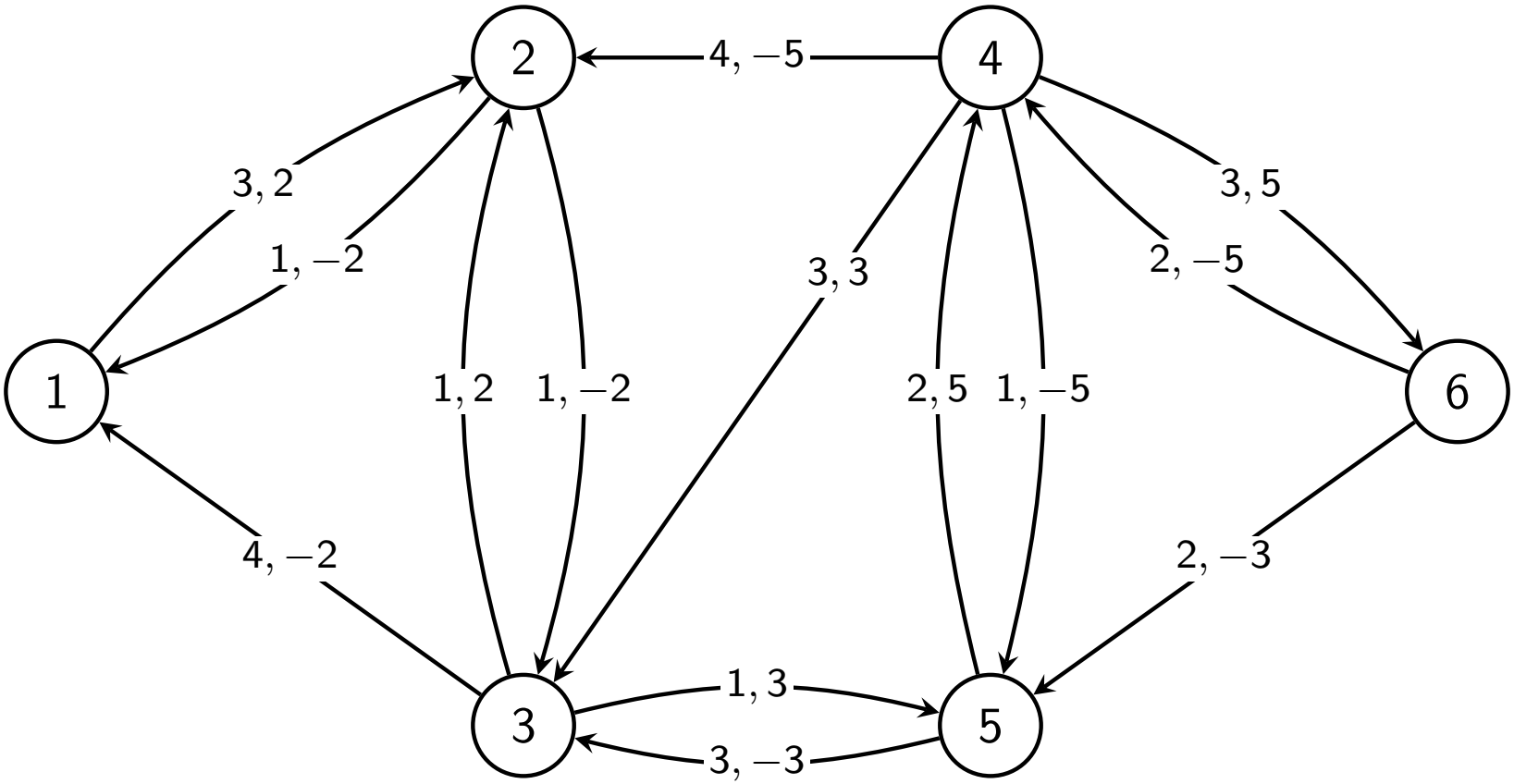


Example: iteration 3 (1/5)

- In $G(\boldsymbol{x}^2)$ the arc $(5, 6)$ is saturated: only its backward arc $(6, 5)$ remains, with $r = 2$.
- The arcs $(6, 4)$, $(4, 5)$ have residual capacities 2 and 1.
- The forward arcs $(4, 6)$, $(5, 4)$ have residual capacities 3 and 2.

Example: iteration 3 (2/5)

$G(\boldsymbol{x}^2)$, with the pair r_{ij}, c_{ij}^x on each arc:



Example: iteration 3 (3/5)

Bellman–Ford on $G(\boldsymbol{x}^2)$ gives:

k	$d^k(1)$	$d^k(2)$	$d^k(3)$	$d^k(4)$	$d^k(5)$	$d^k(6)$
0	0	0	0	0	0	0
1	-2 (2)	-5 (4)	-3 (5)	-5 (6)	-5 (4)	0
2	-7 (2)	-10 (4)	-8 (5)	-5	-10 (4)	0
3	-12 (2)	-10	-13 (5)	-5	-10	0
4	-15 (3)	-11 (3)	-13	-5	-10	0
5	-15	-13 (1)	-13	-5	-10	0
6	-15	-13	-15 (2)	-5	-10	0

Example: iteration 3 (4/5)

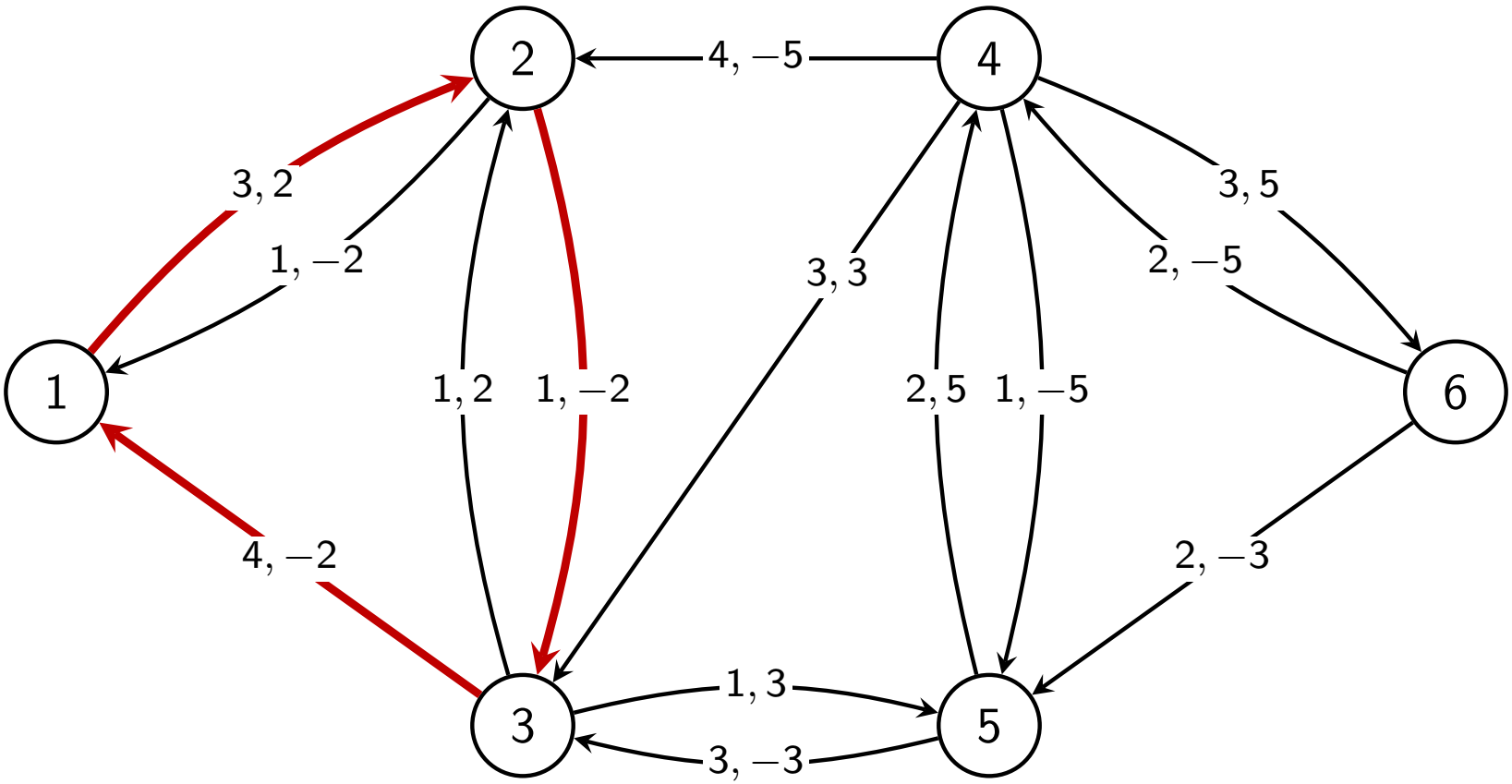
- Only node 3 decreases in pass 6: $j = 3$, and the walk is $6, 4, 5, 3, 1, 2, 3$.
- Node 3 is met twice: $W = 3, 1, 2, 3$, made of the backward arc of $(1, 3)$, the forward arc $(1, 2)$ and the backward arc of $(3, 2)$:

$$c(W) = -2 + 2 - 2 = -2, \qquad r(W) = \min\{4, 3, 1\} = 1.$$

- The new flow is $\boldsymbol{x}^3 = (2, 3, 4, 0, 3, 0, 2, 1, 2) = \tilde{\boldsymbol{x}}$, of cost 60.

Example: iteration 3 (5/5)

The cycle $W = 3, 1, 2, 3$ in $G(\boldsymbol{x}^2)$ is red: augmenting by $r(W) = 1$ empties the arc $(3, 2)$.



Example: iteration 4

- Bellman–Ford on $G(\tilde{\boldsymbol{x}})$ repeats the rows 0–5 of iteration 3, and in pass 6 no label decreases:

$$d^6 = d^5 = (-15, -13, -13, -5, -10, 0).$$

- The algorithm stops with $\tilde{\boldsymbol{x}}$, certified by

$$\boldsymbol{\pi} = -d^6 = (15, 13, 13, 5, 10, 0),$$

already normalized with $\pi_6 = 0$: the potentials of the chapter «Network flow models».

The iterations of cycle canceling

iteration	cost of $\boldsymbol{x}$	j	cycle W	$c(W)$	$r(W)$
1	80	1	$4, 5, 3, 4$	-11	1
2	69	1	$6, 4, 5, 6$	-7	1
3	62	3	$3, 1, 2, 3$	-2	1
4	60	–	no negative cycle	–	–

- The bound of Theorem 3 allows $80 - 60 = 20$ iterations; three are enough.
- The three cycles are the cycles of the decomposition of the example «from the most expensive flow to the optimal flow», found in a different order.

Summary

The cycle-canceling algorithm:

- It starts from a feasible flow and cancels negative cycles of $G(\boldsymbol{x})$
- The cycles are found by Bellman–Ford from labels all equal to 0

Correctness:

- With integer data every cancellation lowers the cost by at least 1
- At the stop the labels give the potentials that certify the optimal flow
- The number of iterations is pseudo-polynomial

The successive shortest path algorithm

Introduction: The successive shortest path algorithm

- The cycle-canceling algorithm keeps the flow feasible and looks for the certificate
- In this section we do the opposite: we keep the certificate true and build the flow, sending it along shortest paths from excess nodes to deficit nodes
- The potentials make all the lengths non-negative, so Dijkstra's algorithm can be used

Definition 5: pseudoflow and imbalances

*A **pseudoflow** is a vector $\boldsymbol{x}$ with $\boldsymbol{l} \leq \boldsymbol{x} \leq \boldsymbol{u}$, which may violate flow conservation. The **imbalance** of node i is*

$$e_i = b_i - \left(\sum_{(i,j)\in\delta^+(i)} x_{ij} - \sum_{(j,i)\in\delta^-(i)} x_{ji} \right) :$$

*the supply that i has still to send if $e_i > 0$ (an **excess node**), the demand that it has still to receive if $e_i < 0$ (a **deficit node**).*

Pseudoflows

- A pseudoflow is a feasible flow exactly when $\boldsymbol{e} = 0$.
- The imbalances add up to zero, since the balances do and every arc adds x_{ij} to its tail and subtracts it from its head.
- The residual network $G(\boldsymbol{x})$ of a pseudoflow is defined as in Definition 1.

Augmenting along a path

- Augmenting $\boldsymbol{x}$ along a path P of $G(\boldsymbol{x})$ from s to t by

$$\delta \leq r(P) = \min_{(i,j)\in P} r_{ij}$$

  is defined as for cycles:
  - the bounds still hold,
  - e_s decreases by δ, e_t increases by δ,
  - the cost grows by $\delta\, c(P)$, where $c(P) = \sum_{(i,j)\in P} c^x_{ij}$.

The idea of the algorithm

- Keep potentials with non-negative reduced costs on $G(\boldsymbol{x})$, the condition of Theorem 2, while the imbalances go to zero.
- With $\boldsymbol{x} = 0$, $\boldsymbol{\pi} = 0$ and costs $c_{ij} \geq 0$ the condition holds at the start.

Next: how to update the potentials after an augmentation so that the reduced costs stay non-negative.

Lemma 3: shortest paths keep the reduced costs non-negative (1/2)

Let $\boldsymbol{x}$ *be a pseudoflow and* $\boldsymbol{\pi}$ *potentials with*

$$\bar{c}_{ij}^{\pi} = c_{ij}^{x} - \pi_i + \pi_j \geq 0 \quad \textit{on every arc of } G(\boldsymbol{x}).$$

Let $d(i)$ *be the distances from a node* s *in* $G(\boldsymbol{x})$ *with the lengths* $\bar{c}^{\pi}$ ($d(i) = +\infty$ *if* i *is not reachable*), t *a node with* $d(t) < +\infty$, P *a shortest path from* s *to* t, *and*

$$\pi'_i = \pi_i - \min\{d(i), d(t)\}, \qquad \forall i \in N.$$

Lemma: shortest paths keep the reduced costs non-negative (2/2)

(a) $c^x_{ij} - \pi'_i + \pi'_j \geq 0$ *on every arc of* $G(\boldsymbol{x})$.

(b) $c^x_{ij} - \pi'_i + \pi'_j = 0$ *on every arc of* P.

(c) *If* $\boldsymbol{x}'$ *is* $\boldsymbol{x}$ *augmented along* P *by* $\delta \leq r(P)$, *then* $c^{x'}_{ij} - \pi'_i + \pi'_j \geq 0$ *on every arc of* $G(\boldsymbol{x}')$.

Proof (1/3)

Let $\hat{d}(i) = \min\{d(i), d(t)\}$. The reduced cost of (i, j) with $\boldsymbol{\pi}'$ is

$$\bar{c}_{ij}^{\pi} + \hat{d}(i) - \hat{d}(j).$$

(a) Let (i, j) be an arc of $G(\boldsymbol{x})$.

- If $d(i) < +\infty$, a shortest path to i followed by (i, j) gives $d(j) \leq d(i) + \bar{c}_{ij}^{\pi}$, so

$$\hat{d}(j) \leq \min\{d(i) + \bar{c}_{ij}^{\pi},\, d(t)\} \leq \hat{d}(i) + \bar{c}_{ij}^{\pi},$$

  since $\bar{c}_{ij}^{\pi} \geq 0$.
- If $d(i) = +\infty$, then $\hat{d}(i) = d(t) \geq \hat{d}(j)$.

In both cases the new reduced cost is non-negative.

Proof (2/3)

(b) Subpaths of P are shortest (Lemma «optimality of subpaths» of the chapter «Shortest paths»), so

$$d(j) = d(i) + \bar{c}_{ij}^{\pi} \qquad \text{on every arc of } P,$$

and $d(i) \leq d(t)$ along P since the lengths are non-negative.

Hence $\hat{d} = d$ on P, and the new reduced cost is 0.

Proof (3/3)

(c) Augmenting along P changes $G(\boldsymbol{x})$ only on the arcs of P:

- an arc (i, j) of P may disappear, and the reverse arc (j, i) may appear;
- the reverse of a residual arc has the opposite cost, hence by (b) reduced cost 0;
- the other arcs keep their reduced costs, non-negative by (a).

□

Why «successive shortest paths»

- For every path P from s to t the potentials telescope:

$$\sum_{(i,j)\in P} \bar{c}_{ij}^{\pi} = c(P) - \pi_s + \pi_t.$$

- A shortest path for the reduced costs is a minimum cost path of $G(\boldsymbol{x})$.

- The flow travels along cheapest paths, hence the name of the algorithm.

Algorithm: Successive shortest path algorithm (1/2)

Input : a balanced network with $\boldsymbol{l} = 0$, integer b_i and u_{ij}, costs $c_{ij} \geq 0$

Output : an optimal flow $\boldsymbol{x}$ with its certificate $\boldsymbol{\pi}$, or «infeasible»

1 **Step 0 (initialization).** Set $\boldsymbol{x} := 0$, $\boldsymbol{\pi} := 0$ and $\boldsymbol{e} := \boldsymbol{b}$.

2 **Step 1 (selection).** If $\boldsymbol{e} = 0$, `STOP`: $\boldsymbol{x}$ is optimal, with $\boldsymbol{\pi}$.

3 Otherwise let s be the node of smallest index with $e_s > 0$.

Algorithm: Successive shortest path algorithm (2/2)

4 **Step 2 (shortest paths).** Run Dijkstra from s on $G(\boldsymbol{x})$, lengths $c^x_{ij} - \pi_i + \pi_j$.

5 If no node with $e_t < 0$ is reachable, STOP: infeasible.

6 Otherwise t is the reachable node with $e_t < 0$ and smallest $d(t)$, and P its path.

7 **Step 3 (augmentation).** Set $\pi_i := \pi_i - \min\{d(i), d(t)\}$ for every $i \in N$.

8 Augment $\boldsymbol{x}$ along P by $\delta := \min\{e_s, -e_t, r(P)\}$, and update e_s, e_t.

9 Go to Step 1.

Rules of this chapter

- Dijkstra's algorithm breaks ties by the smallest index, as in the chapter «Shortest paths».
- Among the deficit nodes at the same distance, t is the one of smallest index.

Theorem 4: correctness of the successive shortest path algorithm

With the input of the algorithm, at the beginning of every iteration $\boldsymbol{x}$ *is an integer pseudoflow and* $c_{ij}^{x} - \pi_i + \pi_j \geq 0$ *on every arc of* $G(\boldsymbol{x})$*. The algorithm stops after at most*

$$D = \sum_{i:b_i>0} b_i$$

iterations. If it stops in Step 1, $\boldsymbol{x}$ *is an optimal flow; if it stops in Step 2, the network has no feasible flow.*

Proof (1/3)

We prove the invariant, then the bound and the two stops.

1. *Invariant.*
   - At the start $G(0)$ has the forward arcs only, with costs $c_{ij} \geq 0$, and $\boldsymbol{\pi} = 0$.
   - Dijkstra's algorithm applies, the lengths being non-negative, and Lemma 3(c) keeps the reduced costs non-negative.
   - The amount δ is a positive integer, since e_s, $-e_t$ and the residual capacities are positive integers (or $+\infty$), so $\boldsymbol{x}$ stays integer.

Proof (2/3)

2. *Bound.*
   - If $e_s > 0$, some node has $e_t < 0$, since the imbalances add up to zero.
   - Every iteration lowers $\sum_{i:e_i>0} e_i$ by $\delta \geq 1$, and no imbalance changes sign, since $\delta \leq e_s$ and $\delta \leq -e_t$.
   - The sum starts at D.

3. *Stop in Step 1.* $\boldsymbol{e} = 0$, so $\boldsymbol{x}$ is a feasible flow, optimal by the invariant and Theorem 2.

Proof (3/3)

4. *Stop in Step 2.* Let S be the set of the nodes reachable from s. No arc of $G(\boldsymbol{x})$ leaves S, so $x_{ij} = u_{ij}$ on $\delta^+(S)$ and $x_{ij} = 0$ on $\delta^-(S)$. The pseudoflow satisfies conservation with the balances $b_i - e_i$, so by Lemma «flow across a cut» of the chapter «Network flow models»

$$\begin{aligned}\sum_{i \in S} b_i &= x(\delta^+(S)) - x(\delta^-(S)) + \sum_{i \in S} e_i \\ &= u(\delta^+(S)) + \sum_{i \in S} e_i > u(\delta^+(S)),\end{aligned}$$

because $e_s > 0$ and no node of S has $e_i < 0$. The cut condition of the chapter «Network flow models» fails for S: there is no feasible flow.

□

Running time and negative costs

- **Running time.** Each iteration runs Dijkstra's algorithm in $O(n^2)$ time: $O(D\,n^2)$ in total, a pseudo-polynomial bound.
- **Negative costs.** If some arcs with finite capacity have $c_{ij} < 0$, start from $x_{ij} = u_{ij}$ on them and $x_{ij} = 0$ elsewhere, with $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{B}\boldsymbol{x}$:
  - such an arc gives only its backward arc, of cost $-c_{ij} > 0$,
  - and $\boldsymbol{\pi} = 0$ works again.

Example: successive shortest paths

- We start from

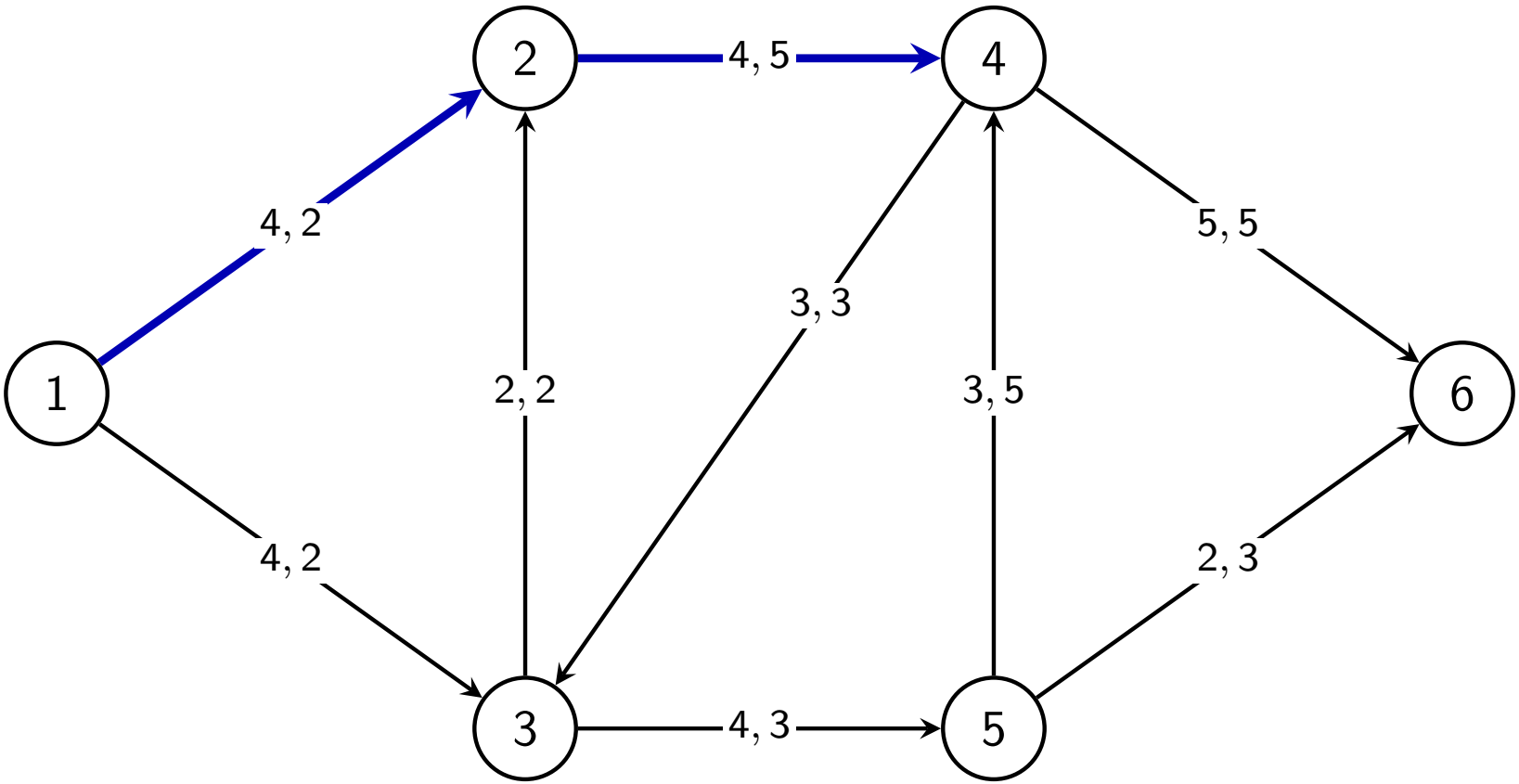
$$\boldsymbol{x} = 0, \qquad \boldsymbol{\pi} = 0, \qquad \boldsymbol{e} = \boldsymbol{b} = (5, 2, 0, -3, 0, -4).$$

- In each iteration we show $G(\boldsymbol{x})$ with r_{ij} and the reduced cost $c^x_{ij} - \pi_i + \pi_j$ on each arc, and the path P in blue.

- Then: the node s, the distances of Dijkstra's algorithm, the node t, the path P with its reduced cost $d(t)$ and its cost $c(P)$, the amount δ, and the new potentials, imbalances and flow.

Example: iteration 1 (1/2)

$G(0)$ has the forward arcs only; the reduced costs are the costs.
Path $P = 1, 2, 4$ in blue:



Example: iteration 1 (2/2)

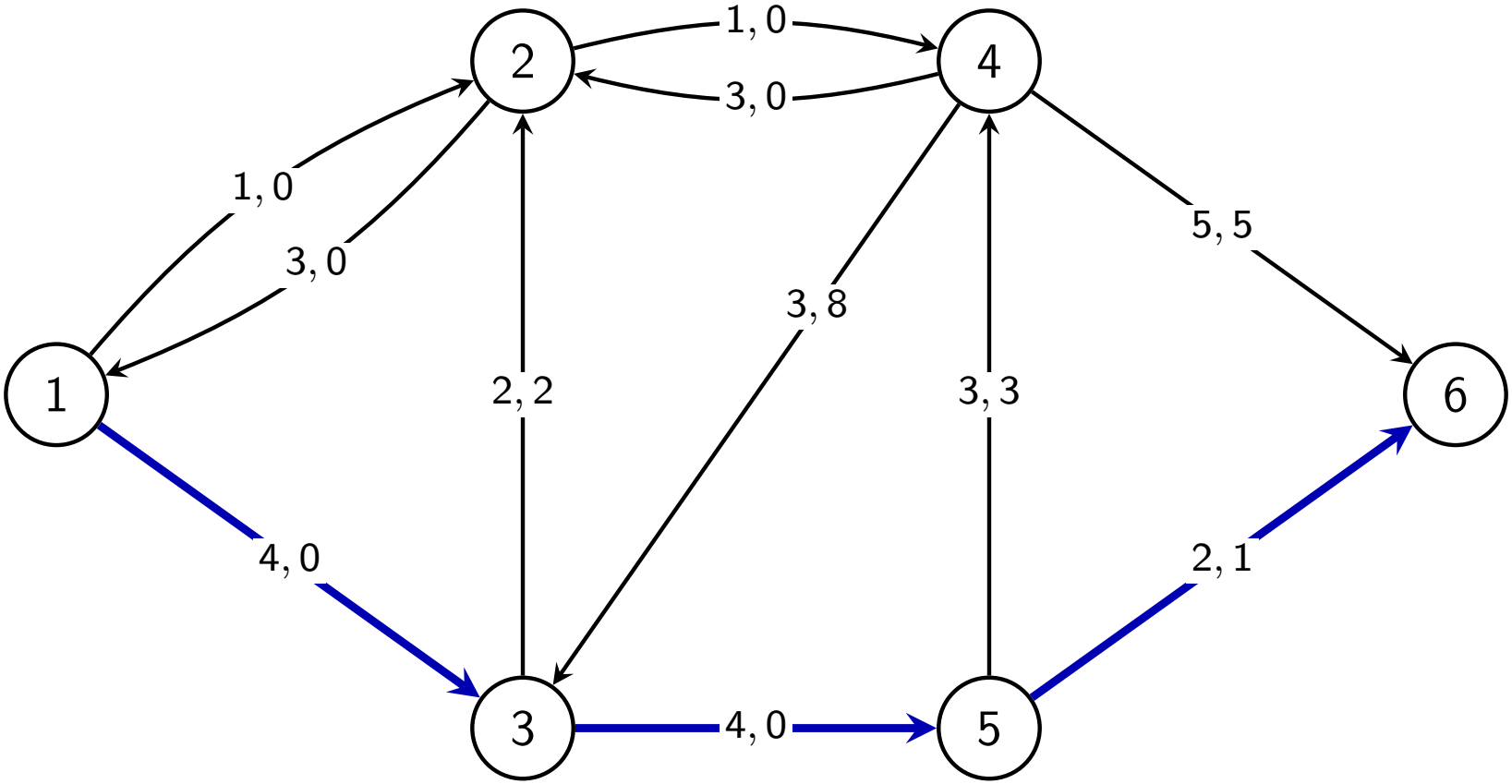
- $s = 1$, and Dijkstra gives $d = (0, 2, 2, 7, 5, 8)$.
- The deficit nodes 4 and 6 are at distance 7 and 8: $t = 4$, and $P = 1, 2, 4$, with $d(t) = 7$ and $c(P) = 7$.
- $\delta = \min\{5, 3, 4, 4\} = 3$, and the new flow costs 21.
- Node 6 is farther than t and its potential decreases by 7, not 8.

After the iteration:

$$\boldsymbol{\pi} = (0, -2, -2, -7, -5, -7),$$
$$\boldsymbol{e} = (2, 2, 0, 0, 0, -4),$$
$$\boldsymbol{x} = (3, 0, 3, 0, 0, 0, 0, 0, 0).$$

Example: iteration 2 (1/2)

$G(\boldsymbol{x})$ with r_{ij} and the reduced cost on each arc. Path $P = 1, 3, 5, 6$ in blue:



Example: iteration 2 (2/2)

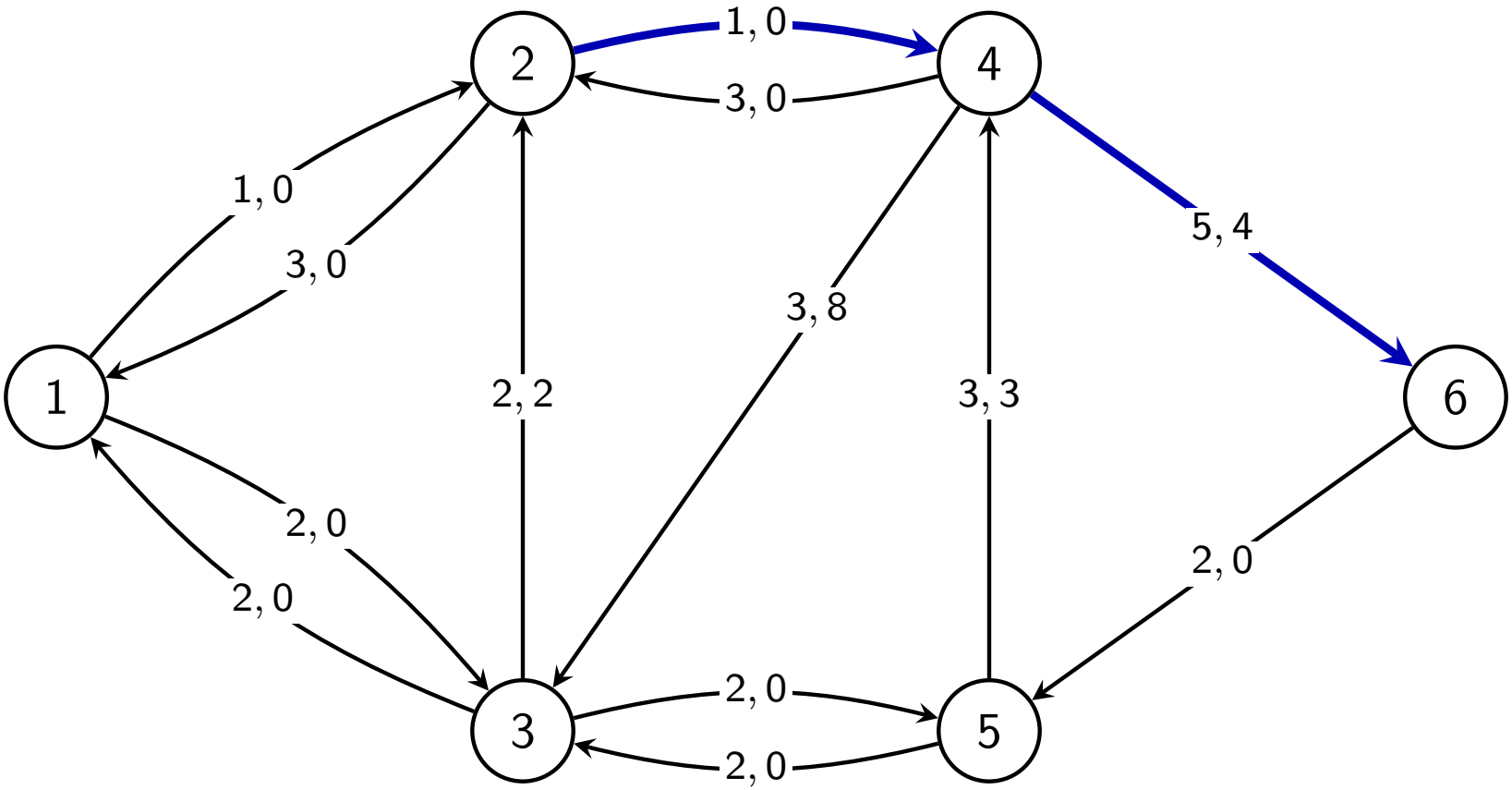
- The arcs $(1,2)$, $(2,4)$ of the first path now have reduced cost 0, and so do their backward arcs.
- $s = 1$, and Dijkstra gives $d = (0,0,0,0,0,1)$: $t = 6$.
- The path $1,3,5,6$ has reduced cost $d(t) = 1$ and cost $c(P) = 8$.
- $\delta = \min\{2,4,4,4,2\} = 2$ is set by e_1 and by u_{56}; the new flow costs 37.

After the iteration:

$$\boldsymbol{\pi} = (0,-2,-2,-7,-5,-8),$$
$$\boldsymbol{e} = (0,2,0,0,0,-2),$$
$$\boldsymbol{x} = (3,2,3,0,2,0,0,0,2).$$

Example: iteration 3 (1/2)

$G(\boldsymbol{x})$ with r_{ij} and the reduced cost on each arc. Path $P = 2, 4, 6$ in blue:



Example: iteration 3 (2/2)

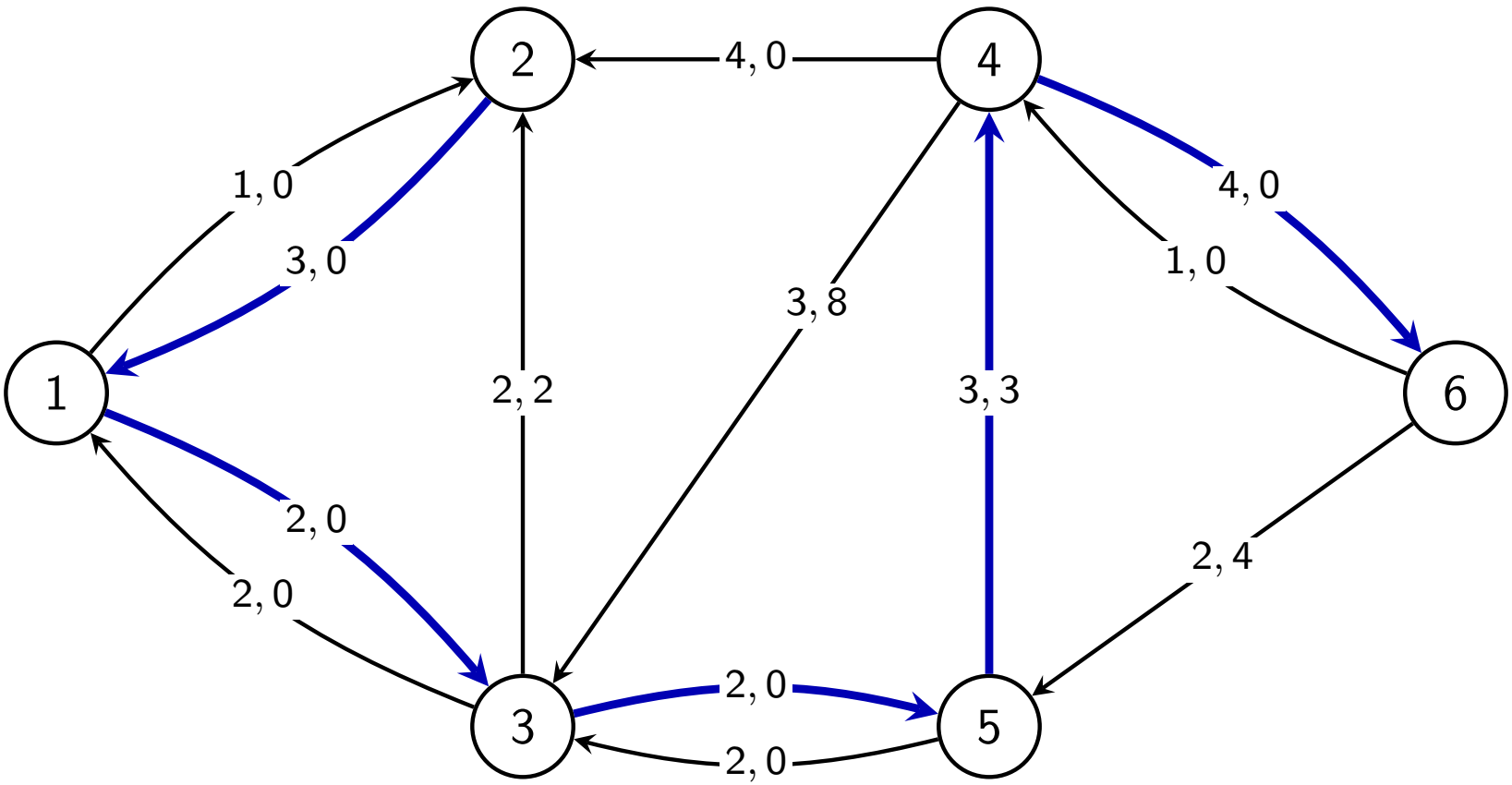
- $s = 2$, and $d(1) = 0$ through the backward arc $(2, 1)$.
- Dijkstra gives $d = (0, 0, 0, 0, 0, 4)$: $t = 6$.
- The path $2, 4, 6$ has reduced cost $d(t) = 4$ and cost $c(P) = 10$.
- It uses the last unit of $(2, 4)$: $\delta = 1$, and the new flow costs 47.

After the iteration:

$$\boldsymbol{\pi} = (0, -2, -2, -7, -5, -12),$$
$$\boldsymbol{e} = (0, 1, 0, 0, 0, -1),$$
$$\boldsymbol{x} = (3, 2, 4, 0, 2, 0, 1, 0, 2).$$

Example: iteration 4 (1/2)

$G(\boldsymbol{x})$ with r_{ij} and the reduced cost on each arc. Path $P = 2, 1, 3, 5, 4, 6$ in blue:



Example: iteration 4 (2/2)

- $s = 2$, and Dijkstra gives $d = (0, 0, 0, 3, 0, 3)$: $t = 6$.
- The path $2, 1, 3, 5, 4, 6$ has reduced cost $d(t) = 3$ and cost $c(P) = 13$; $\delta = 1$, and the new flow costs 60.
- It uses the backward arc $(2, 1)$: one unit that went from 1 to 2 is rerouted from 1 through 3, 5 and 4.

After the iteration:

$$\boldsymbol{\pi} = (0, -2, -2, -10, -5, -15),$$
$$\boldsymbol{e} = (0, 0, 0, 0, 0, 0),$$
$$\boldsymbol{x} = (2, 3, 4, 0, 3, 0, 2, 1, 2).$$

Example: the end of the algorithm

- After iteration 4 the imbalances are zero and $\boldsymbol{x} = \tilde{\boldsymbol{x}}$, of cost 60.
- Shifted by 15, the potentials become

$$(15, 13, 13, 5, 10, 0) :$$

the potentials of the chapter «Network flow models».

Next: the four iterations collected in tables, as in the notes.

Example: the iterations in tables (1/4)

Each row is an iteration, with the node s, the distances of Dijkstra's algorithm and the node t:

it.	s	$d(1)$	$d(2)$	$d(3)$	$d(4)$	$d(5)$	$d(6)$	t
1	1	0	2	2	7	5	8	4
2	1	0	0	0	0	0	1	6
3	2	0	0	0	0	0	4	6
4	2	0	0	0	3	0	3	6

Example: the iterations in tables (2/4)

The path P, its reduced cost $d(t)$ and its cost $c(P)$, the amount δ and the cost of the new flow:

it.	P	$d(t)$	$c(P)$	δ	cost
1	$1, 2, 4$	7	7	3	21
2	$1, 3, 5, 6$	1	8	2	37
3	$2, 4, 6$	4	10	1	47
4	$2, 1, 3, 5, 4, 6$	3	13	1	60

Example: the iterations in tables (3/4)

After each iteration, the potentials, the imbalances and the flow:

it.	π	e	x
0	$(0,0,0,0,0,0)$	$(5,2,0,-3,0,-4)$	$(0,0,0,0,0,0,0,0,0)$
1	$(0,-2,-2,-7,-5,-7)$	$(2,2,0,0,0,-4)$	$(3,0,3,0,0,0,0,0,0)$
2	$(0,-2,-2,-7,-5,-8)$	$(0,2,0,0,0,-2)$	$(3,2,3,0,2,0,0,0,2)$
3	$(0,-2,-2,-7,-5,-12)$	$(0,1,0,0,0,-1)$	$(3,2,4,0,2,0,1,0,2)$
4	$(0,-2,-2,-10,-5,-15)$	$(0,0,0,0,0,0)$	$(2,3,4,0,3,0,2,1,2)$

Example: the iterations in tables (4/4)

The reduced costs $\bar{c}_{ij} = c_{ij} - \pi_i + \pi_j$ of the arcs of A, before each iteration (the forward arc has $\bar{c}_{ij}$ if $x_{ij} < u_{ij}$, the backward arc $-\bar{c}_{ij}$ if $x_{ij} > 0$):

it.	$(1,2)$	$(1,3)$	$(2,4)$	$(3,2)$	$(3,5)$	$(4,3)$	$(4,6)$	$(5,4)$	$(5,6)$
1	2	2	5	2	3	3	5	5	3
2	0	0	0	2	0	8	5	3	1
3	0	0	0	2	0	8	4	3	0
4	0	0	0	2	0	8	0	3	-4

The costs of the paths grow

- Here the costs $7, 8, 10, 13$ of the paths grow from one iteration to the next: later units travel along more expensive paths, and the total cost is

$$3 \cdot 7 + 2 \cdot 8 + 10 + 13 = 60.$$

- Four iterations are enough here, while the bound of Theorem 4 is $D = 7$.

Observation 1: Dijkstra needs the reduced costs

In iteration 4 *the arcs* $(2,1)$, $(3,1)$, $(4,2)$, $(5,3)$, $(6,4)$, $(6,5)$ *of* $G(\boldsymbol{x})$ *are backward arcs with negative costs* c^x_{ij}, *and Dijkstra's algorithm with negative lengths can fail (chapter «Shortest paths»).*

The potentials make all the lengths non-negative without changing the shortest paths.

Summary

The successive shortest path algorithm:

- It sends flow from an excess node to a deficit node along a shortest path of $G(\boldsymbol{x})$
- The path is found by Dijkstra's algorithm on the reduced costs

Correctness:

- The update $\pi_i := \pi_i - \min\{d(i), d(t)\}$ keeps the reduced costs non-negative
- The final flow is optimal and the potentials certify it
- At most D iterations with integer data

Typical mistakes

Introduction: Typical mistakes

- In this section we list the most common mistakes with the residual network and the two algorithms
- They concern the costs of the backward arcs, the size of the augmentations, the use of Bellman–Ford and of Dijkstra, and the potentials
- Recognizing them helps to run the algorithms correctly by hand

Observation 2: typical mistakes (1/3)

- ***Giving the backward arc the cost*** $+c_{ij}$***.*** *A backward arc decreases* x_{ij}*, so it saves* c_{ij} *per unit: its cost is* $-c_{ij}$*, and its residual capacity is* $x_{ij} - l_{ij}$*.*
- ***Augmenting by more than*** $r(W)$***.*** *Some arc of the cycle would leave its bounds;* $\theta = r(W)$ *is the largest step.*
- ***Running Bellman–Ford from a single node.*** *A negative cycle that is not reachable from that node is missed; the labels start at* 0 *on every node.*

Observation: typical mistakes (2/3)

- ***Taking the walk for the cycle.*** *The walk with n arcs read from the table contains a negative cycle, but it is not a cycle itself: the cycle is the piece between two occurrences of the same node.*
- ***Running Dijkstra's algorithm on the costs*** c^x_{ij}***.*** *Backward arcs can have negative costs; the lengths are the reduced costs* $c^x_{ij} - \pi_i + \pi_j$*.*

Observation: typical mistakes (3/3)

- ***Updating the potentials with the wrong sign.*** *With $\bar{c}_{ij} = c_{ij} - \pi_i + \pi_j$ the update is $\pi_i := \pi_i - \min\{d(i), d(t)\}$; the nodes farther than t decrease by $d(t)$ only.*
- ***Taking the maximum flow of Step 0 as optimal.*** *The auxiliary network has no costs: its maximum flow is feasible, and optimal only by chance.*

Summary

Residual network and cycle canceling:

- Backward arcs cost $-c_{ij}$; augment by at most $r(W)$
- Bellman–Ford starts from labels 0 on every node; the cycle is a piece of the walk

Successive shortest paths:

- Dijkstra runs on the reduced costs, not on the costs
- $\pi_i := \pi_i - \min\{d(i), d(t)\}$, with the minus sign
- The maximum flow of Step 0 is feasible, not optimal in general

In summary

Introduction: In summary

- We collect the key ideas of the chapter in a few points
- We close with the reference network at a glance: the flows, the cycles, the paths and the potentials computed in this chapter

Summary (1/3)

- The residual network $G(\boldsymbol{x})$ has a forward arc (i, j) with $r_{ij} = u_{ij} - x_{ij}$ and cost c_{ij}, and a backward arc (j, i) with $r_{ji} = x_{ij} - l_{ij}$ and cost $-c_{ij}$; augmenting along a cycle W changes the cost by $\theta\, c(W)$.
- A circulation decomposes into cycles; the difference of two feasible flows is a circulation on $G(\boldsymbol{x})$.
- A feasible flow is optimal if and only if $G(\boldsymbol{x})$ has no negative cycle, if and only if some potentials give non-negative reduced costs on $G(\boldsymbol{x})$; on the arcs of A: $\bar{c}_{ij} > 0 \Rightarrow x_{ij} = l_{ij}$, $\bar{c}_{ij} < 0 \Rightarrow x_{ij} = u_{ij}$.

Summary (2/3)

- A maximum flow from a super source to a super sink finds a feasible flow or proves that there is none.
- Cycle canceling: Bellman–Ford from labels 0 finds a negative cycle, augmenting by $r(W)$ cancels it; at most $\boldsymbol{c}'\boldsymbol{x}^0 - z(\mathrm{MCF})$ iterations with integer data.
- Successive shortest paths: Dijkstra on the reduced costs from an excess node to the nearest deficit node, $\pi_i := \pi_i - \min\{d(i), d(t)\}$; at most D iterations.

Summary (3/3)

The reference network at a glance.

most expensive flow	$\boldsymbol{x}^0 = (1, 4, 4, 1, 4, 1, 3, 3, 1)$, cost 80
cycle canceling from $\boldsymbol{x}^0$	cycles $4, 5, 3, 4$; $6, 4, 5, 6$; $3, 1, 2, 3$
	costs $80, 69, 62, 60$
successive shortest paths	paths $1, 2, 4$; $1, 3, 5, 6$; $2, 4, 6$; $2, 1, 3, 5, 4, 6$
	costs $21, 37, 47, 60$
first feasible flow	maximum flow of value 7; on A it is $\tilde{\boldsymbol{x}}$
optimal flow	$\tilde{\boldsymbol{x}} = (2, 3, 4, 0, 3, 0, 2, 1, 2)$, $z(\mathrm{MCF}) = 60$
potentials ($\pi_6 = 0$)	$\boldsymbol{\pi} = (15, 13, 13, 5, 10, 0)$